

PHYSIOLOGY OF THE NEURON

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INTRODUCTION

STRUCTURE OF THE NEURON

AXONAL TRANSPORT

RESTING POTENTIAL

1/Recording

1/ Passive phenomena

2/ Active phenomena

LOCAL (ELECTROTONIC) POTENTIALS

1/ characters

2/ electrical model of the membrane

THE ACTION POTENTIAL

1/Recording

2/ characters

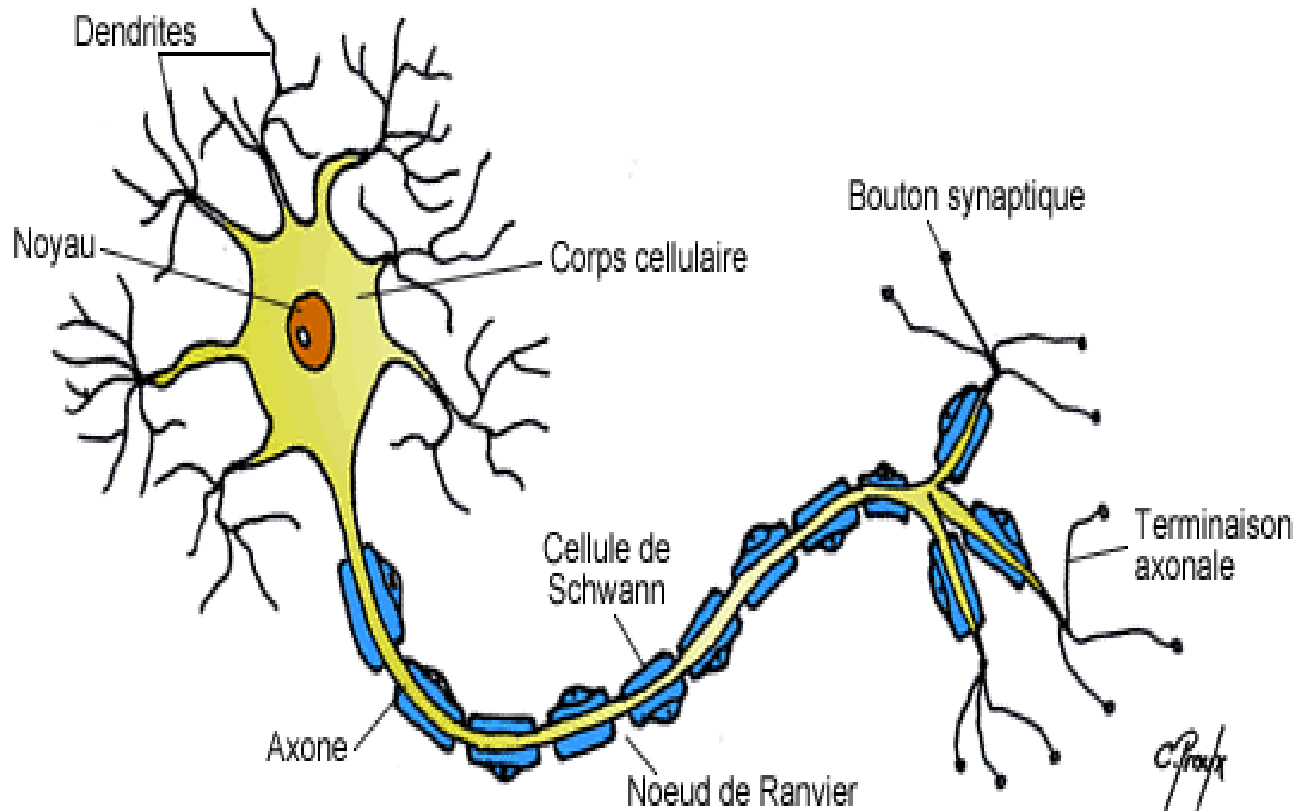
3/ ionic mechanisms

4/ ion channels

NERVE CONDUCTION

EXPLORATION

STRUCTURE OF THE NEURON

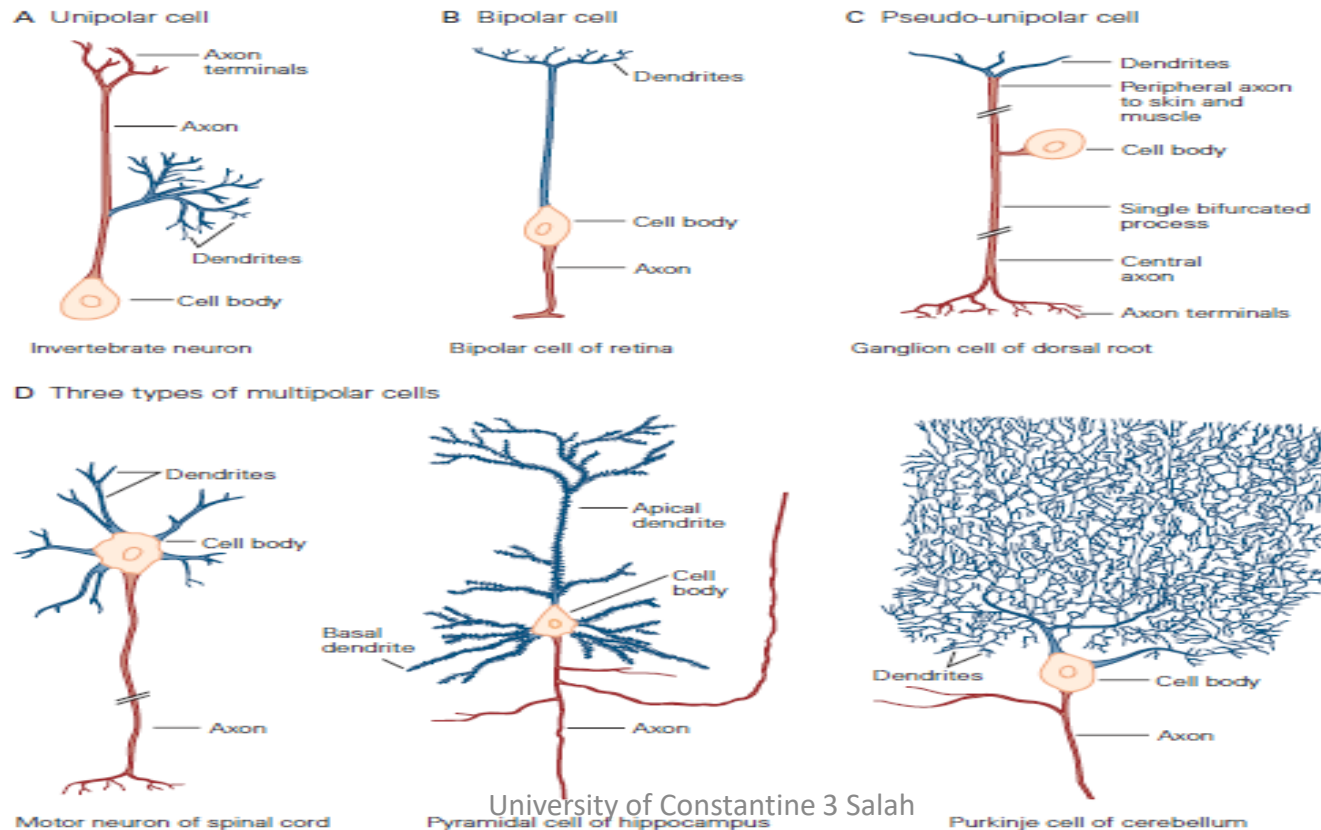


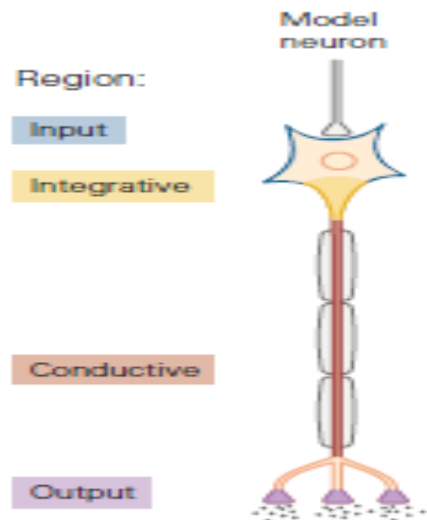
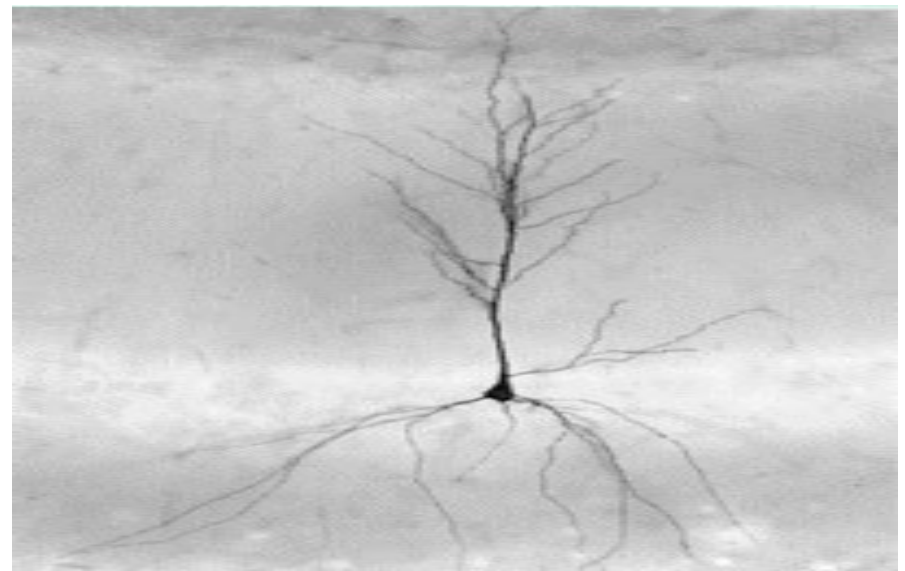
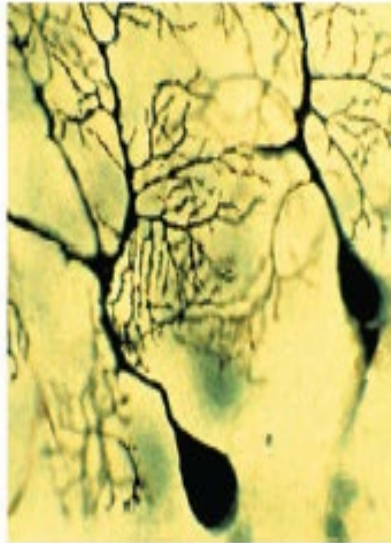
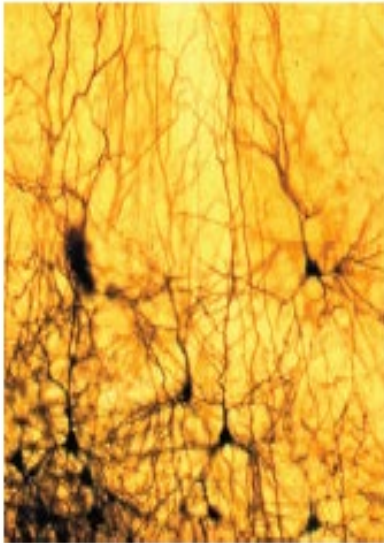
A neuron is made up of:

1-**A soma (perikarion):** contains nucleus and cytoplasm and is of variable morphology pyramidal, ovoid, granular, stellate, fusiform.

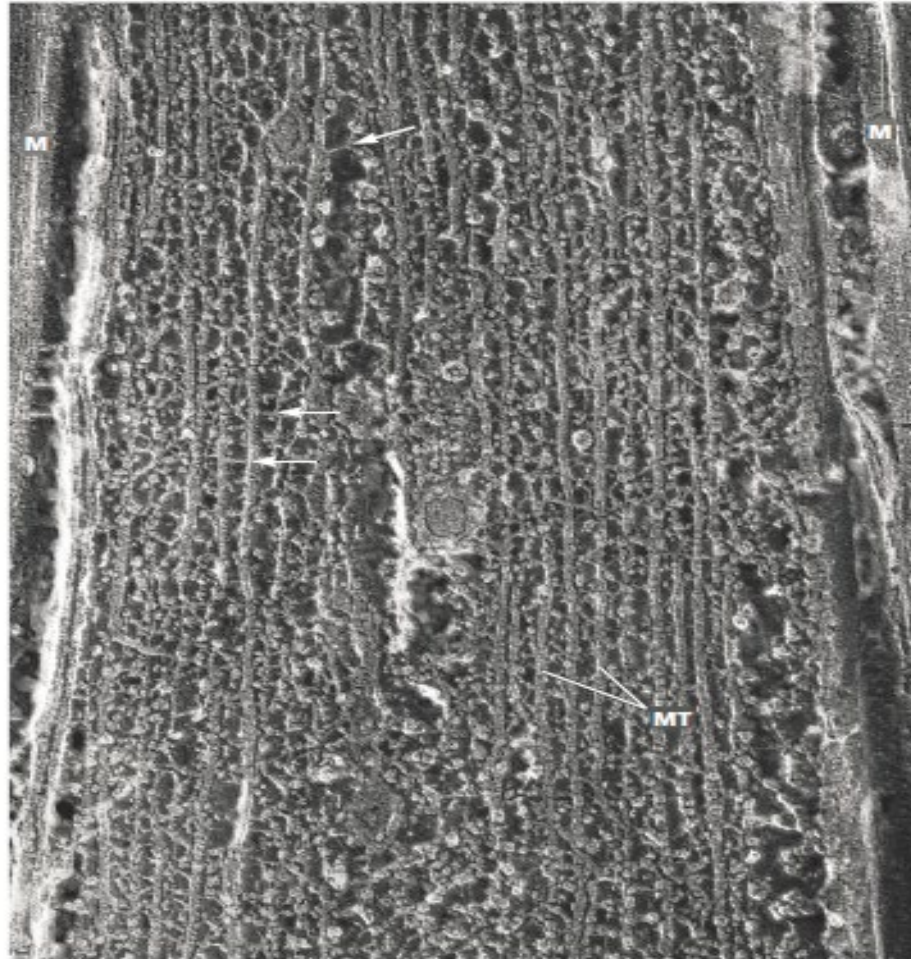
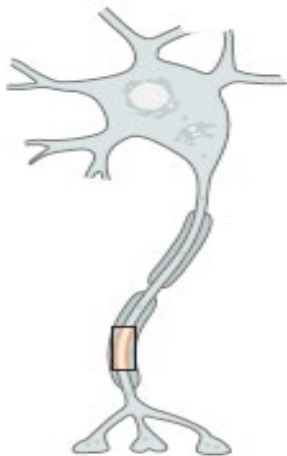
2-**Dendrites: extensions** of the soma, generally spiny (or smooth)

3-**The axon:** non-spiny (smooth), originates at the level of the axon hillock , may have collaterals and may be or not myelinated.





ELEMENTS OF THE CYTOSKELETON



Microtubules (about 24 μm in diameter) are polymers of basic tubulin units.

Microtubules move in cytoplasmic expansions.

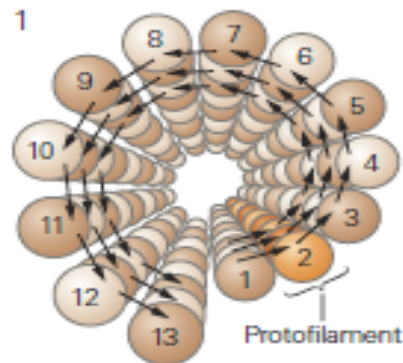
Specific proteins (**MAP**, for microtubule associated proteins) are associated with the tubules which play a role in the transport of other proteins.

Microfilaments : small in size (5 μm in diameter) resulting from a helical assembly of two filaments which are actin polymers. Very numerous in dendrites as in the axon.

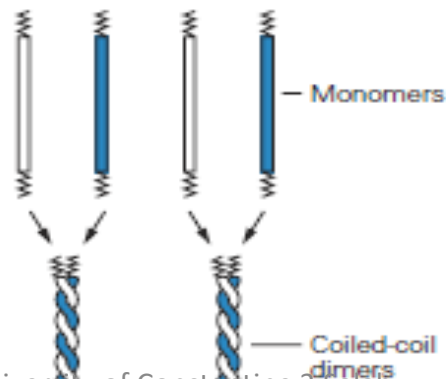
Their attachment to the plasma membrane gives them a role in the rigidity of cytoplasmic expansions.

Neurofilaments : intermediate in size between microtubules and microfilaments (10 μm). Formed of long, highly compressed protein chains, they are more stable and more rigid than other filaments.

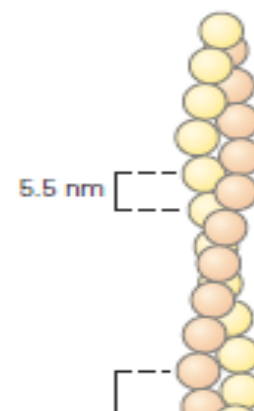
A Microtubule

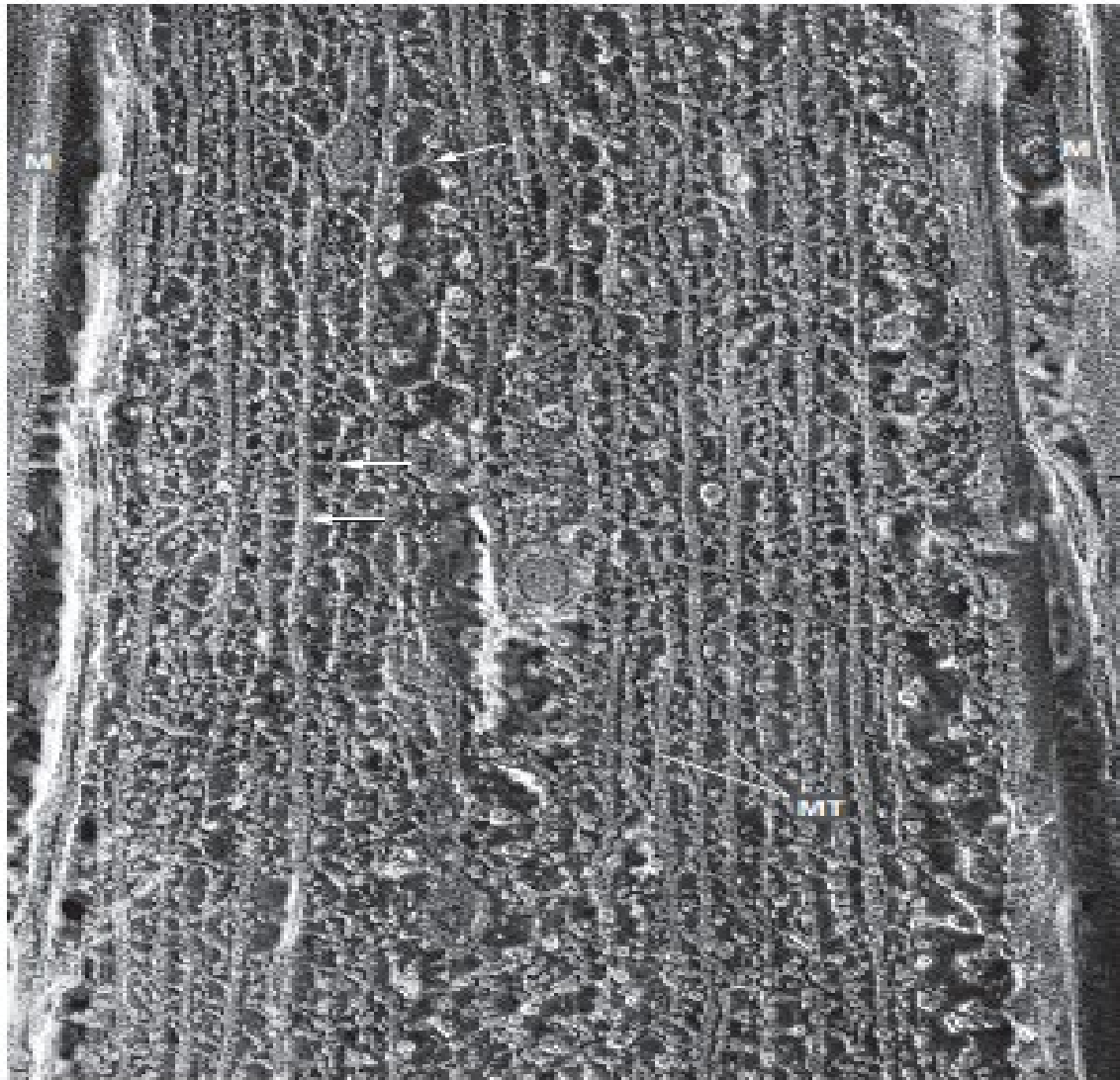
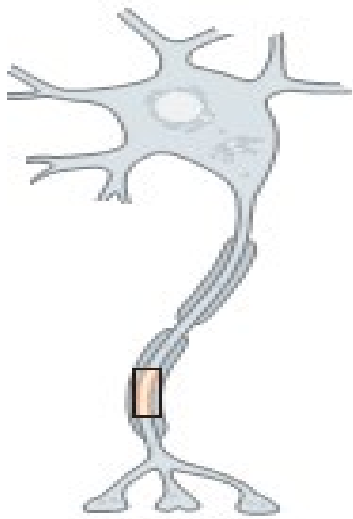


B Neurofilament



C Microfilament

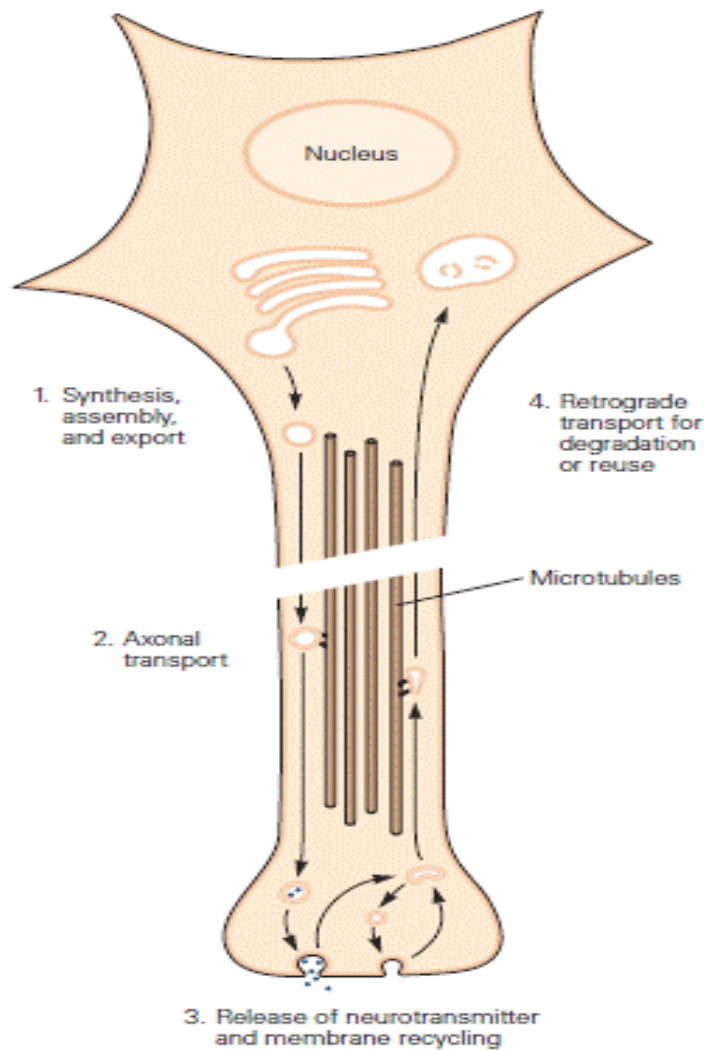




MT: microtubules x 105,000

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AXONAL TRANSPORT



Fast anterograde transport (100-400 mm/day): renewal of axon membrane proteins, NT synthesis enzymes and NT precursors

Slow anterograde transport (0.1-2mm/day):
Renewal of the cytoskeleton, provides the axoplasm of growing axons.

Mitochondria transport :
Renewal of mitochondria in the axon and terminals.

Retrograde axonal transport : role in waste elimination.

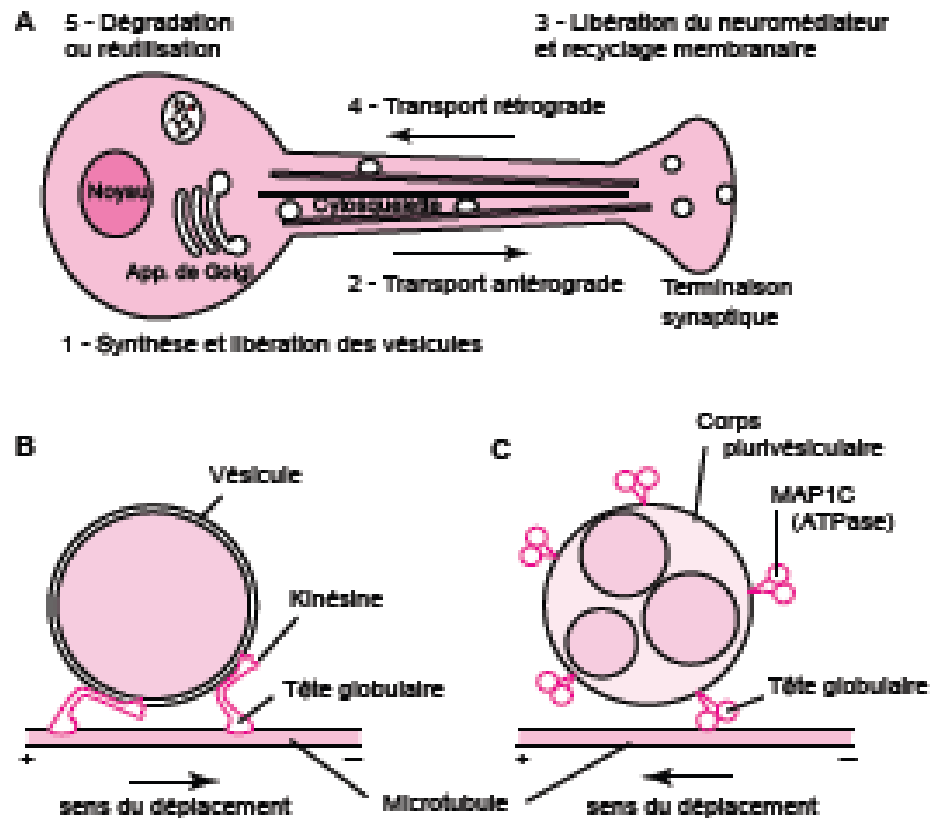


Figure 3 : Transports vésiculaires

A – Schéma d'ensemble des transports antérograde et rétrograde ;

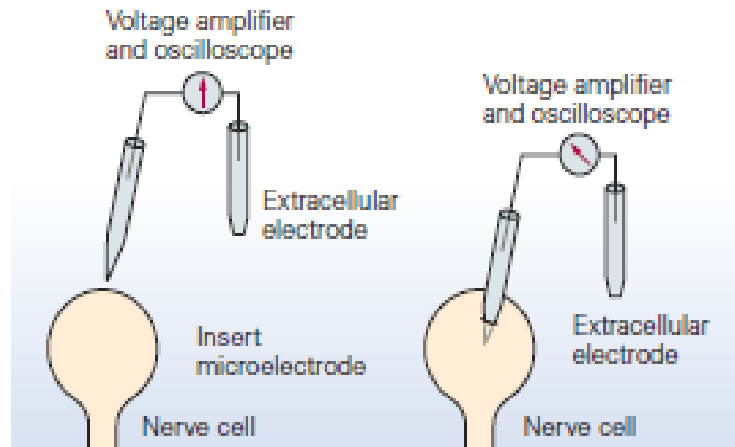
B – Transport antérograde rapide assuré par une ATPase spécifique, la kinésine ;

C – Transport rétrograde assuré par la MAP1C

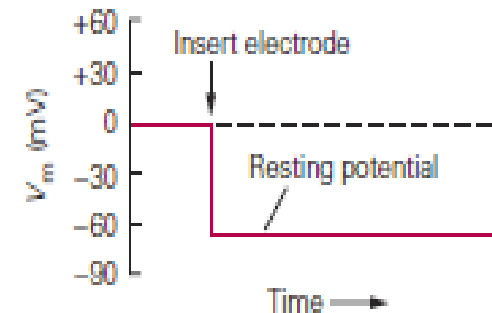
Neural Signaling

RESTING MEMBRANE POTENTIAL

RECORDING OF THE RESTING POTENTIAL:



The recording setup.



Oscilloscope display.

When the microelectrode is in the extracellular space, the voltmeter does not register any difference of potential. If we introduce the end of the microelectrode in the cell, we record a potential difference of about -60mV. This difference of potential, called **resting potential**, is variable from one cell to another and characteristic of all living cells.

“The electrical properties that result from this ddp are at the origin of the functioning of neurons”.

PASSIVE MECHANISMS

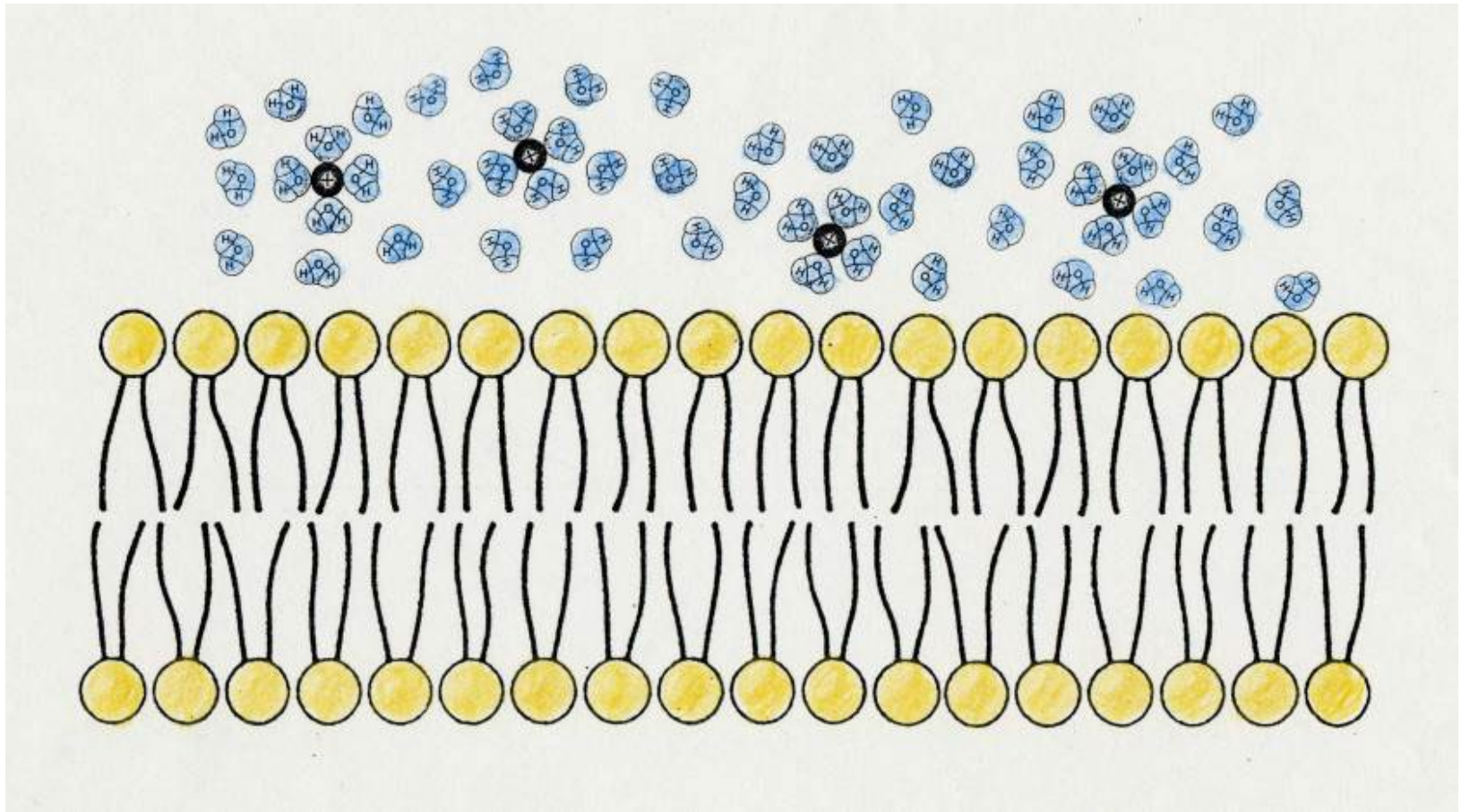
Intracellular and extracellular environments are aqueous and contain substances in solution.

- The interior of the cell is rich in K^+ and A^- (Pr, AA, HCO_3^- ...)
- Externally, Na^+ and Cl^- predominate (see table).

Species of ion	Concentration in cytoplasm (mM)	Concentration in extracellular fluid (mM)
K^+	400	20
Na^+	50	440
Cl^-	52	560
A^- (organic anions)	385	none

Distribution of major ions across the membrane of a squid giant axon

Charged particles cannot pass through the barrier (**hydrophobic**) of the lipid bilayer

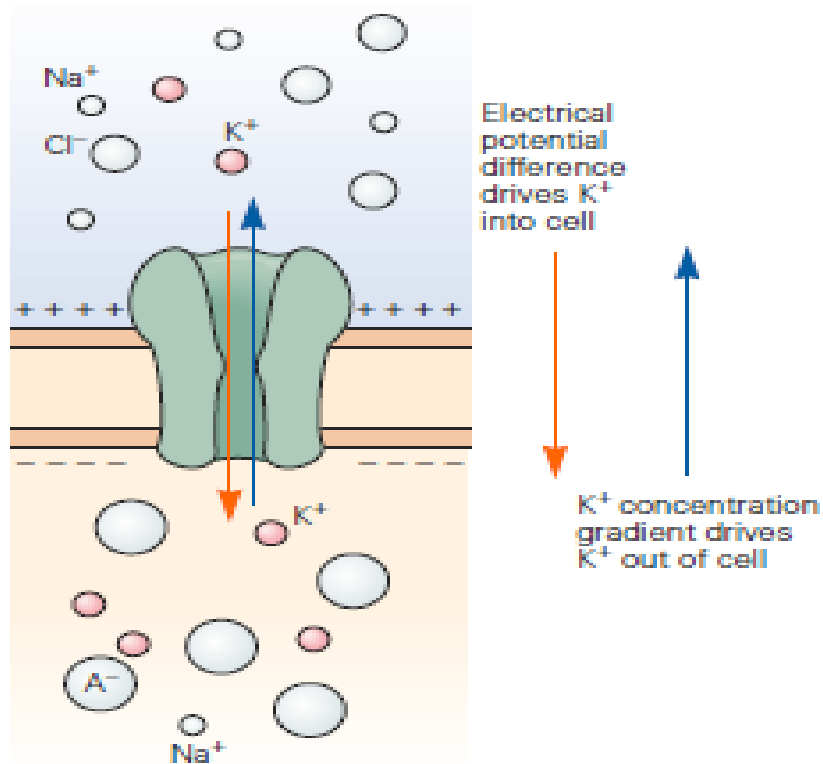


Since electrical charges are carried by ions, the resulting charge separation causes passive movement of ions through transmembrane proteins or "channel proteins" that are selective for each ionic species.

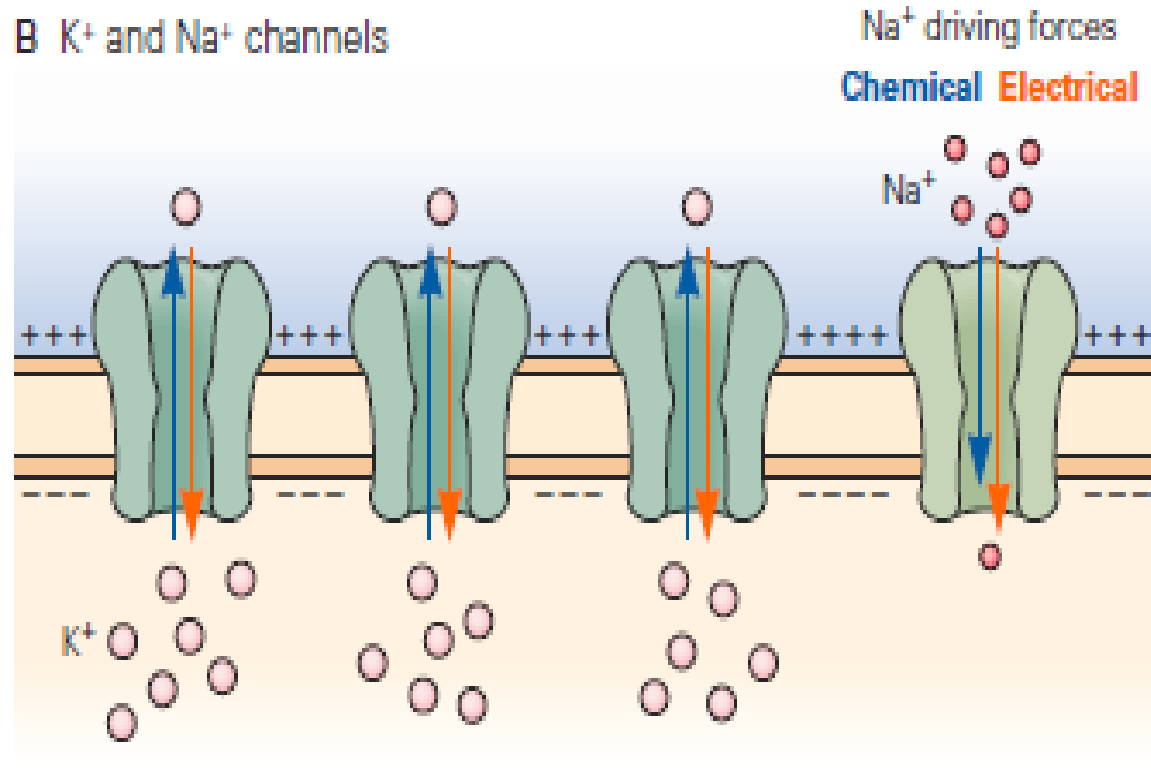
These passive movements are carried out according to two gradients:

- A concentration gradient (osmotic) which tends to equalize the concentrations of the two compartments.
- An electrical gradient due to the resting potential (V_m) and which tends to distribute the ions according to their ionic charge

EXAMPLE OF THE POTASSIUM ION



NET FLOWS OF Na^+ and K^+ IONS



EQUILIBRIUM POTENTIAL: NERNST equation

IF WE CONSIDER A SEMI-PERMEABLE MEMBRANE

At equilibrium, the concentration flux is equal to the electric flux, hence the NERNST equation:

$$E_{\text{ion}} = \frac{R T}{Z F} \ln \frac{[\text{ion}]_e}{[\text{ion}]_i}$$

où: R = constante des gaz parfaits, T = température absolue Z = la valence de l'ion (+ 1 pour le K^+ et le Na^+ , - 1 pour le Cl^- , et +2 pour le Ca^{2+}), F = Faraday, $[\text{ion}]_e$ = concentration de l'ion à l'extérieur de la cellule, $[\text{ion}]_i$ = concentration de l'ion à l'intérieur de la cellule, à 20°C, $RT/F = 25$ mV.

This equation allows us to calculate the equilibrium potential of an ion (E_{ion}), i.e. the membrane potential for which this ion is in equilibrium with respect to the electrochemical forces (zero net flux).

$$E_{\text{Na}^+} = 0,058 \log 440/50 = +55 \text{ mV}$$

$$E_{\text{K}^+} = 0,058 \log 20/400 = -75 \text{ mV}$$

$$E_{\text{Cl}^-} = -0,058 \log 560/50 = -61 \text{ mV}$$

$$E_{\text{Ca}^{2+}} = 0,029 \log 1/0,001 = +116 \text{ mV}.$$

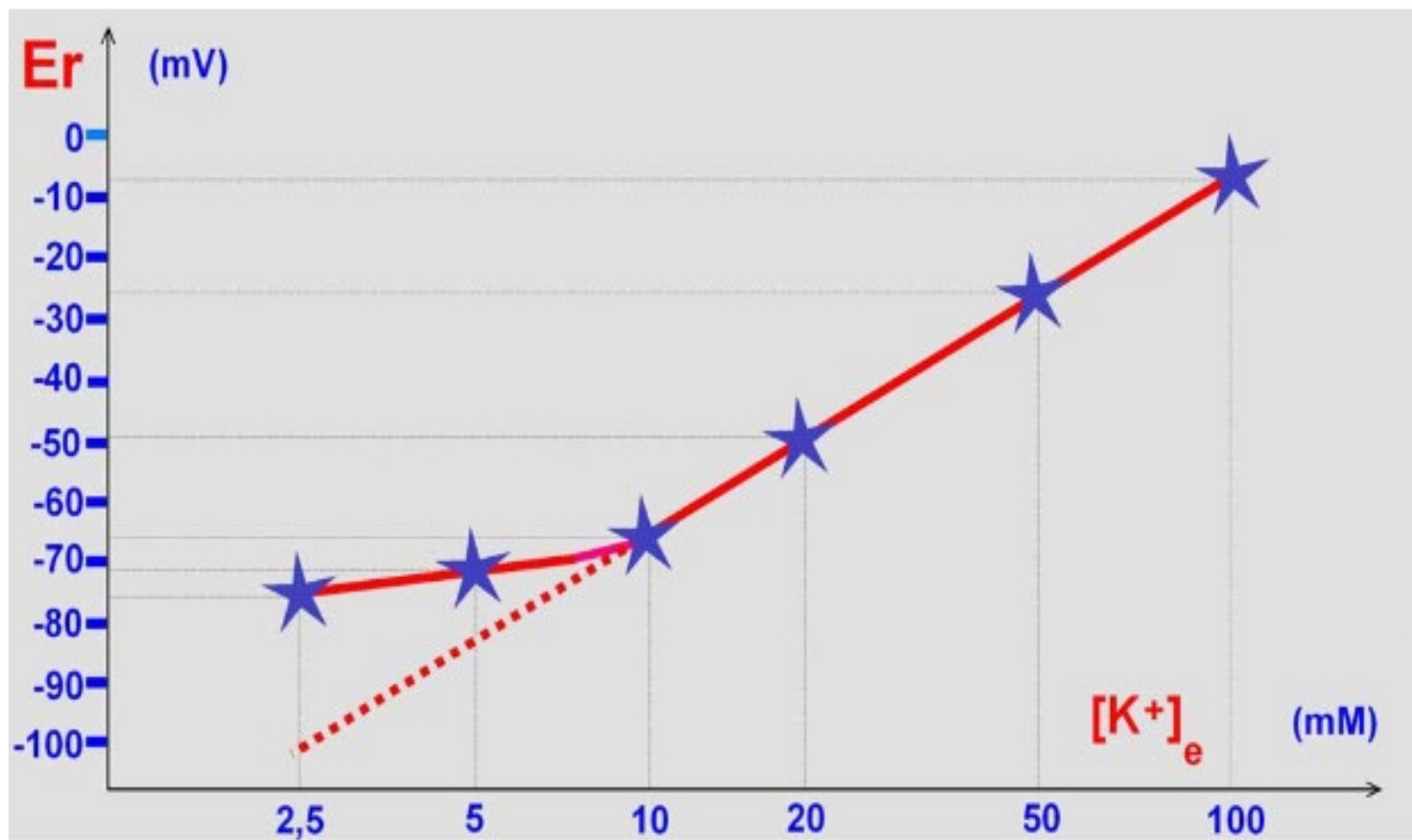


Fig. 3 – Variations des valeurs mesurées de E_r en fonction du $\log [K^+]_e$

MEMBRANE PERMEABILITY: GOLDMAN equation

In reality, the membrane is permeable to several ions and the ioniques fluxes are function not only of electrochemical forces ($E_m - E_{ion}$) but also permeabilities or conductances “**g**” for each ion, it is therefore appropriate to take these permeabilities into account hence the Goldman equation.

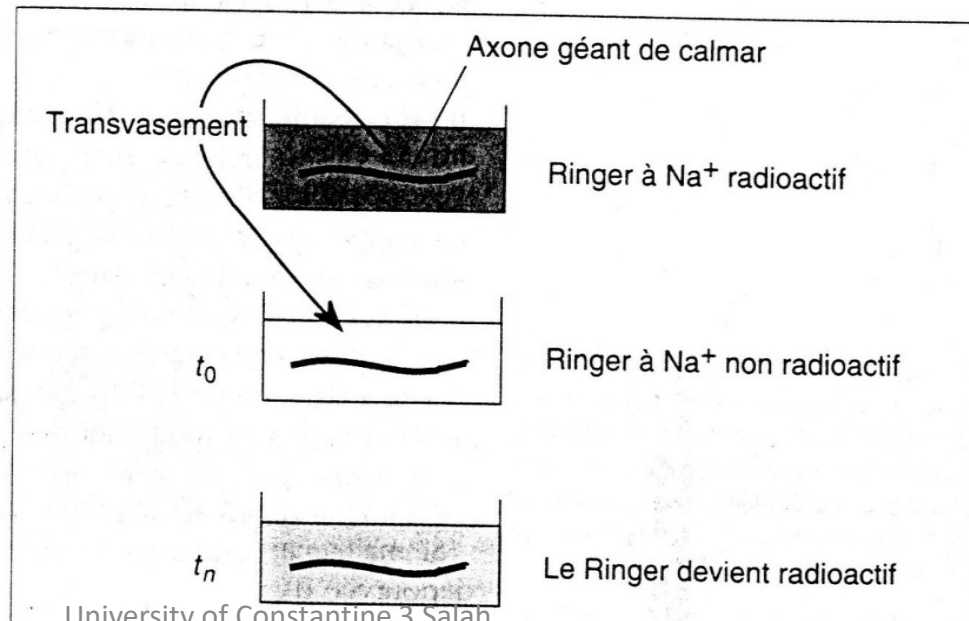
$$V_m = \frac{RT}{F} \ln \frac{P_K [K^+]_o + P_{Na} [Na^+]_o + P_{Cl} [Cl^-]_i}{P_K [K^+]_i + P_{Na} [Na^+]_i + P_{Cl} [Cl^-]_o}$$

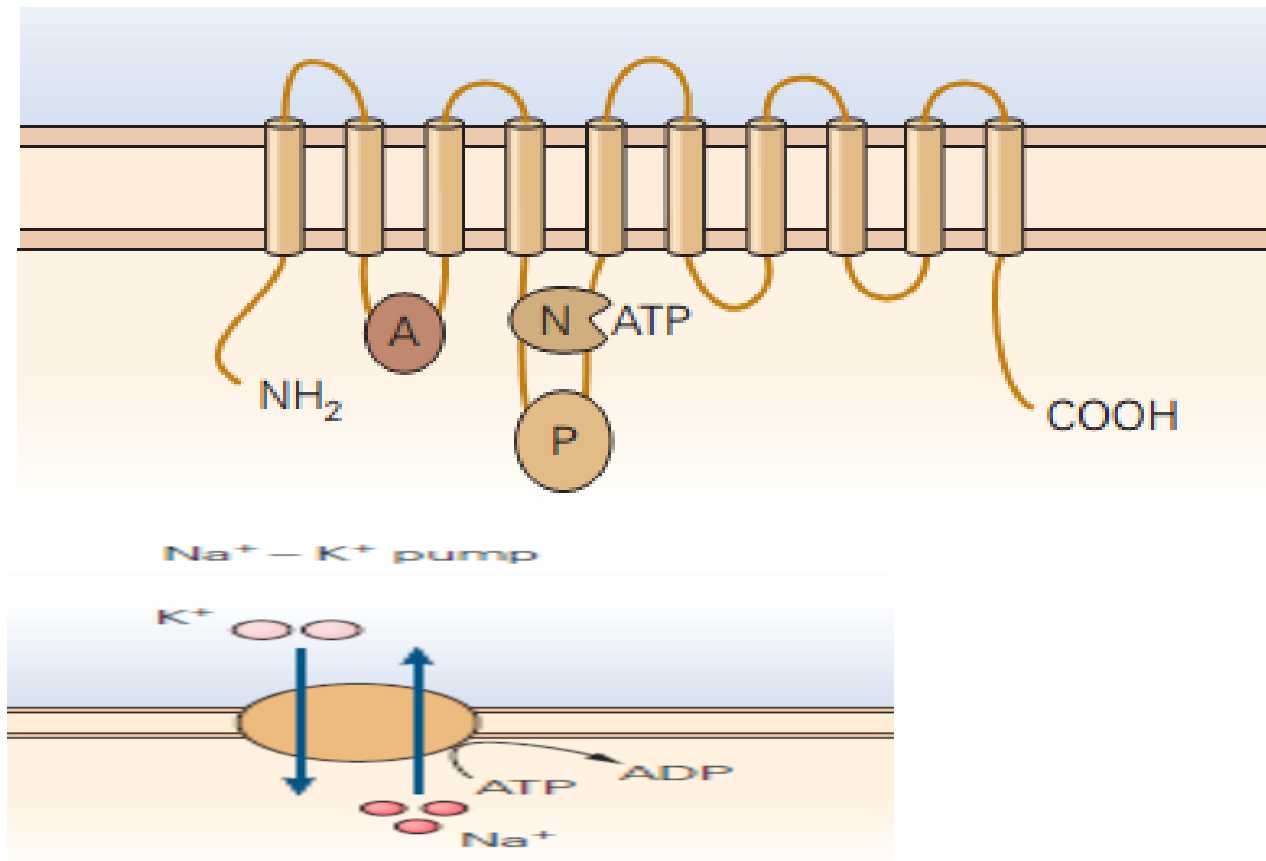
Goldman Equation

ACTIVE MECHANISMS

At rest, there is a permanent flow of Na^+ entering and a permanent flow of K^+ leaving and to ensure the stability of the Na^+ and K^+ concentrations, the intervention of an active process of transport in the opposite direction (against the electrochemical gradients) is necessary: this is the Na^+ - K^+ pump.

Figure 14. Un axone géant de calmar est incubé quelques minutes dans un milieu contenant un isotope de sodium radioactif. L'isotope pénètre passivement dans l'axone selon la force électrochimique. Au temps t_0 , l'axone est placé dans un milieu normal ; au temps t_n le milieu devient progressivement radioactif. La membrane a donc rejeté cet ion de façon active contre sa force électrochimique.





The unequal flux of Na⁺ and K⁺ ions causes the pump to generate a **net outward ionic current**. Thus, the pump is said to be **electrogenic**. This pump-driven efflux of positive charge tends to set the resting potential a few millivolts more negative than would be achieved by the passive diffusion mechanisms.

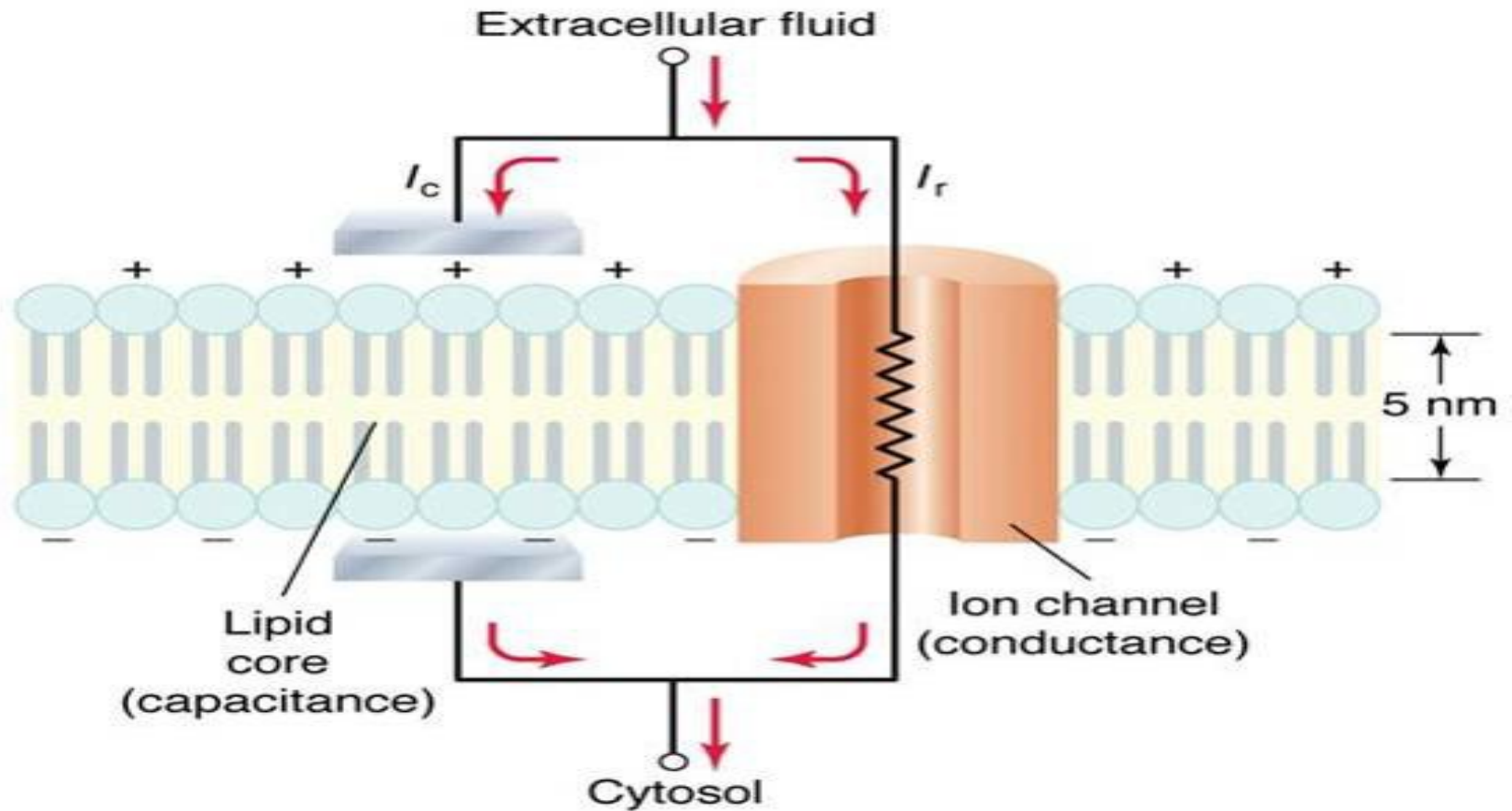
EFFECTS OF SUBTHRESHOLD STIMULATION: LOCAL (Electrotonic) POTENTIALS

Characteristics of local potentials:

- Gradualable
- Summable
- Decremental
- Depolarizing or hyperpolarizing
- Do not respond to the all-or-nothing law

The Passive Electrical Properties of the Neuron affect Electrical signaling

The equivalent circuit of the neuronal membrane



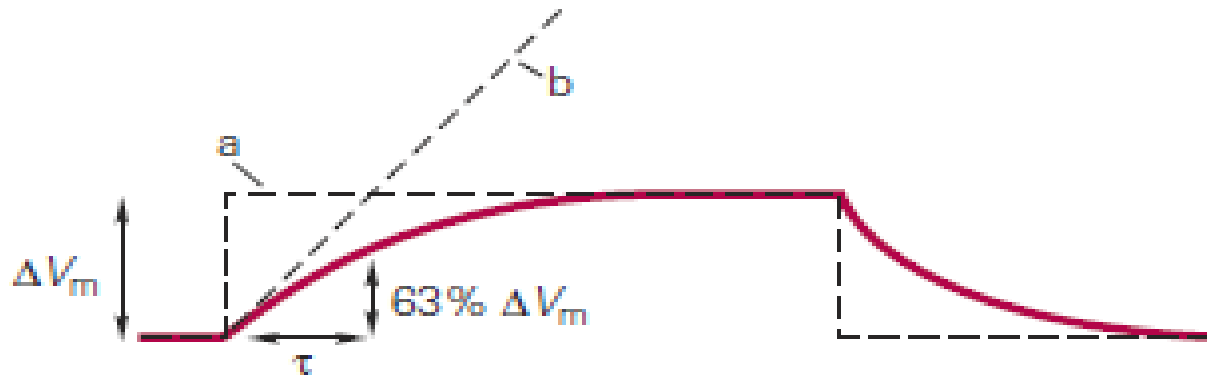
- These local phenomena are due to the passive physical properties of the membrane.
- The lipid bilayer constitutes the equivalent of a capacitor (**C_m**); the conductive proteins oppose a resistance (**R_m**) to the current (**I_m**) crossing the membrane.
- The membrane can therefore be compared to the juxtaposition of elementary circuits connected by longitudinal resistances (**R_L**) of the intracellular compartment.

Two constants are defined :

Local response: time constant .

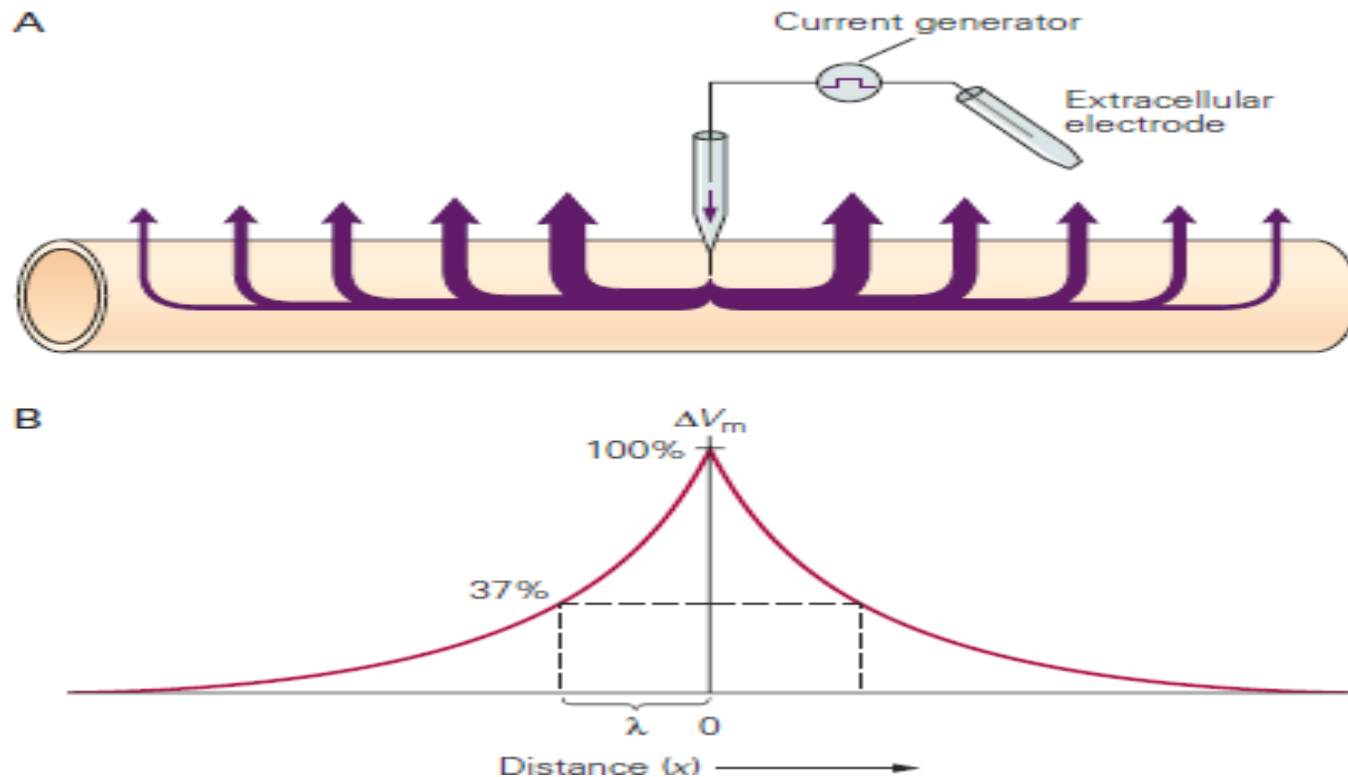
It is a function of the values of C_m .

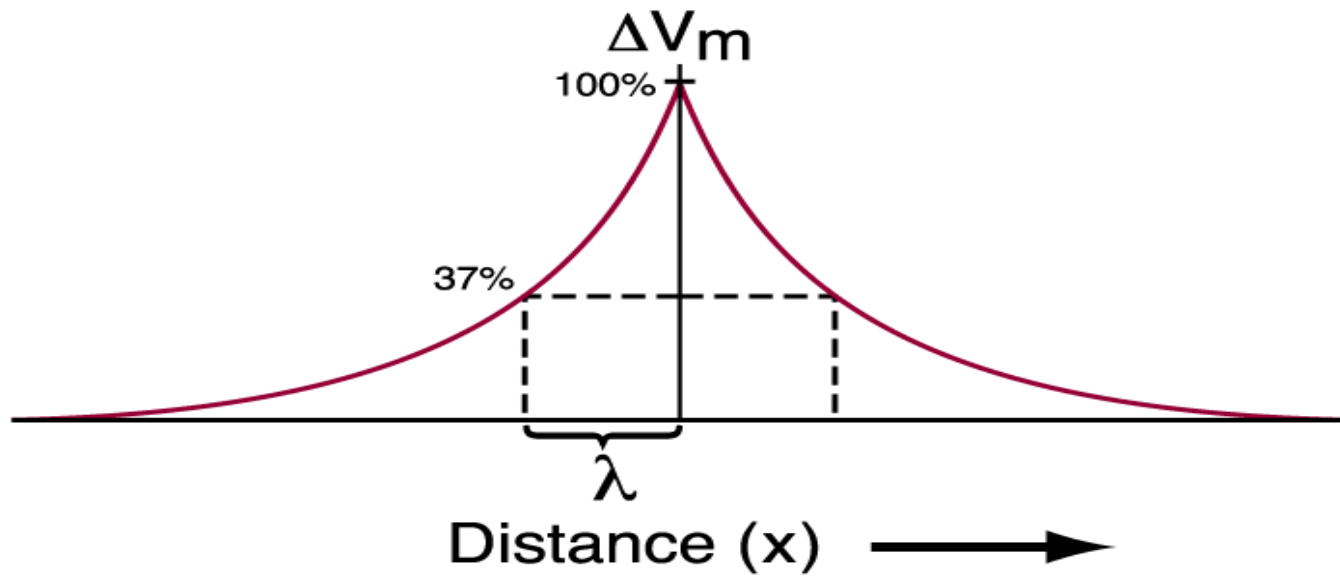
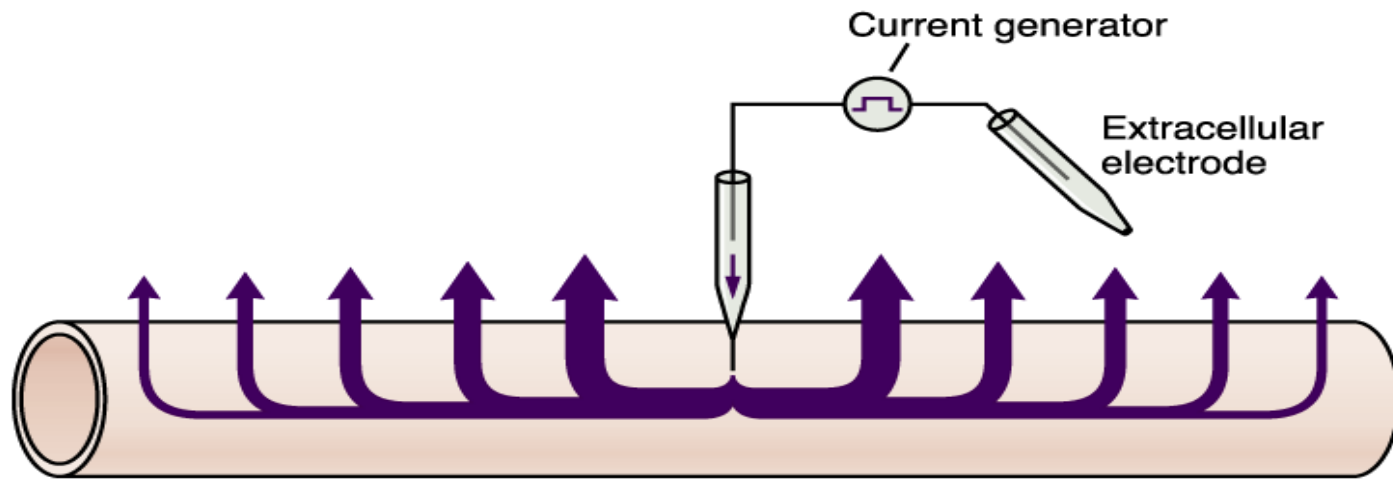
Corresponds to the time required for the ddp to reach 63% of its maximum value.



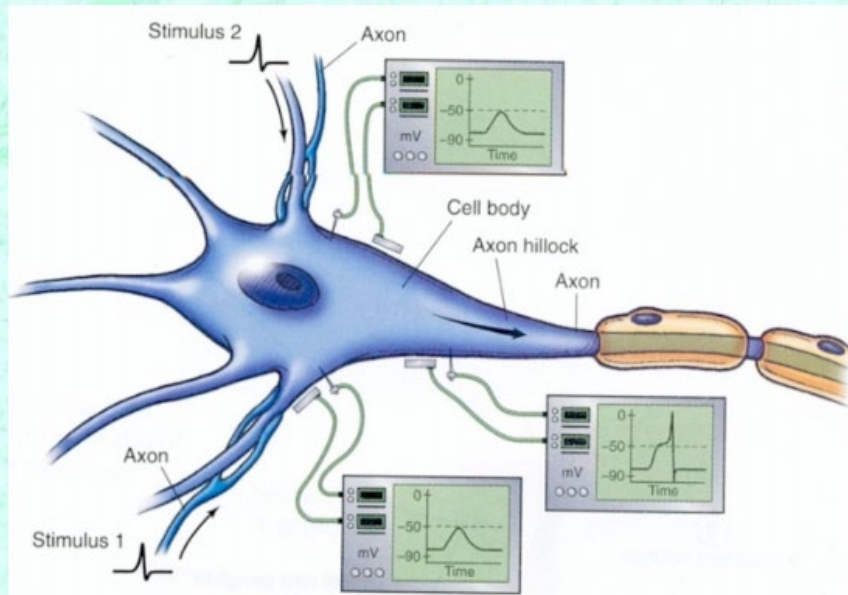
The length constant

The change in V_m decays exponentially with distance from the site of current injection. The distance at which V_m has decayed to 37% of its value at the point of current injection defines the length constant.



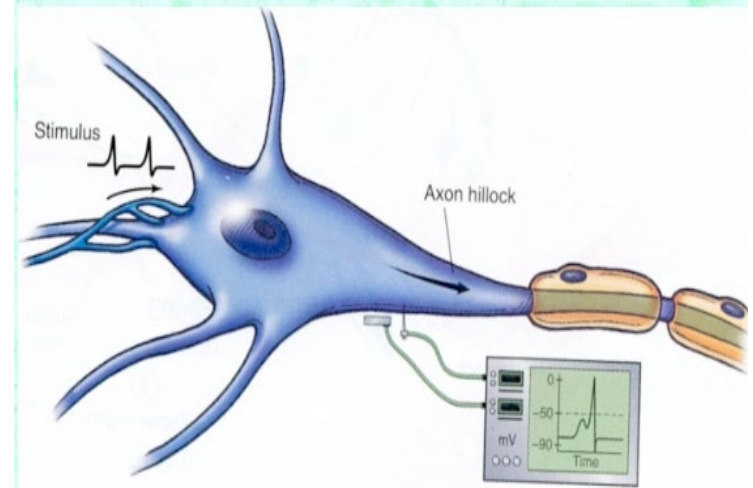


Sommation spatiale



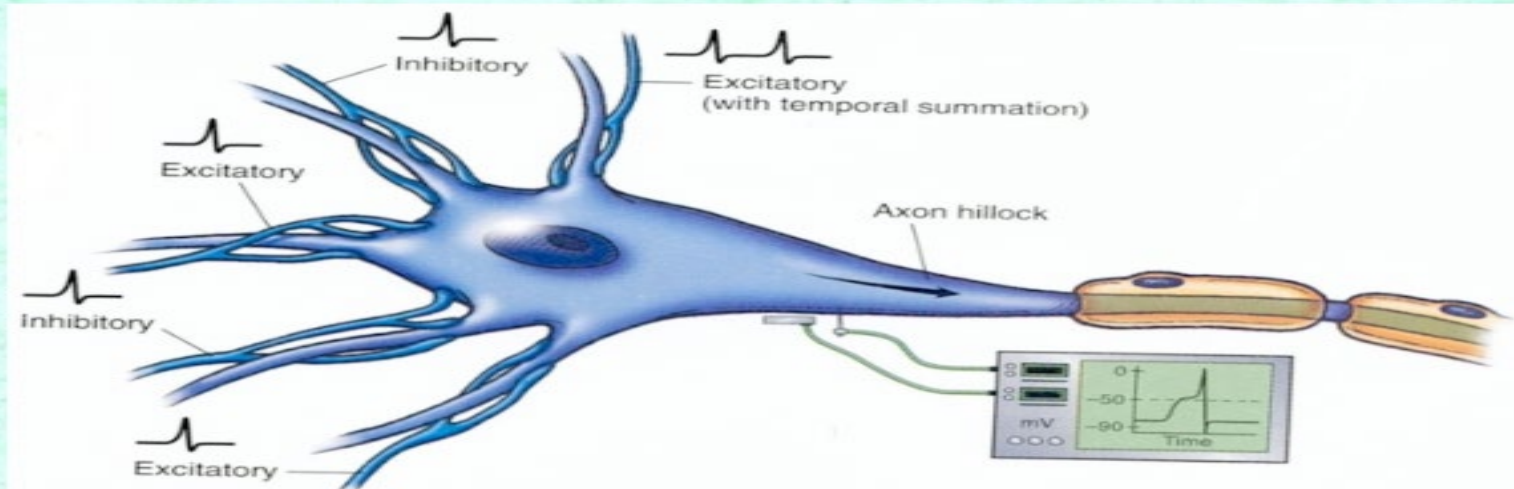
Seeley "Anatomy & Physiology", Mc Graw-Hill

Sommation temporelle



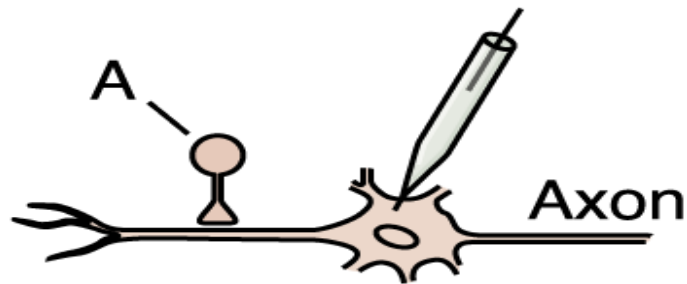
Seeley "Anatomy & Physiology", Mc Graw-Hill

Sommation temporelle et spatiale



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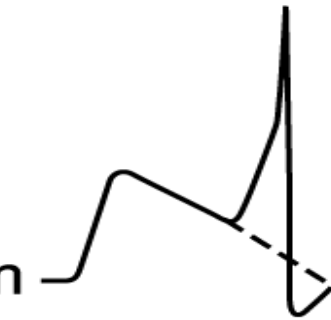
Temporal summation



Synaptic current



Synaptic potential



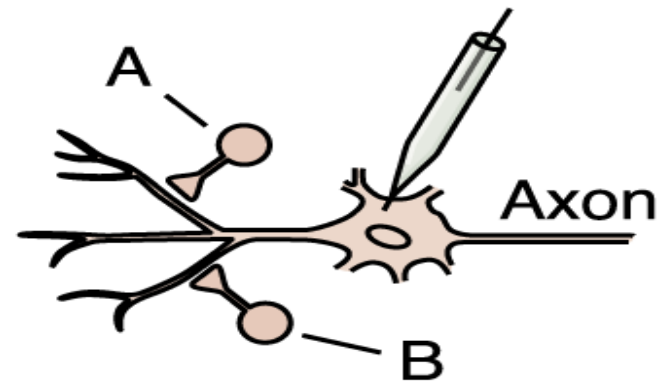
Long time constant (100 ms)



Short time constant (20 ms)

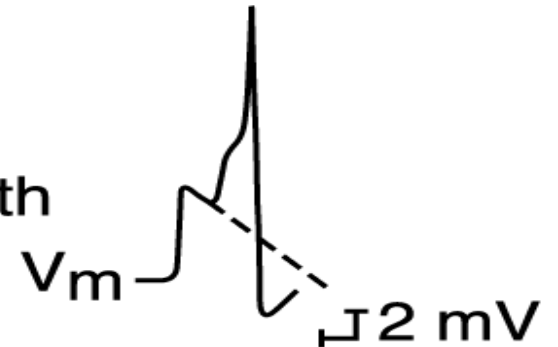


Spatial summation

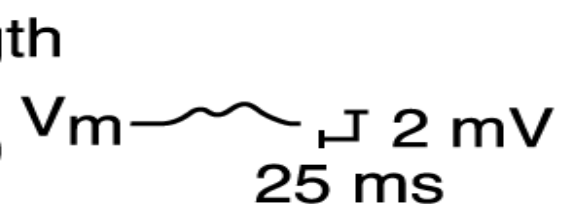


AB

Long length constant (1 mm)



Short length constant (0.33 mm)



ACTION POTENTIAL

Nerve cells are able to carry electrical signals over long distances because the long distance signal, the action potential, is continually regenerated and thus does not attenuate as it moves down the axon.

Premier P.A. (Hodgkin & Huxley)

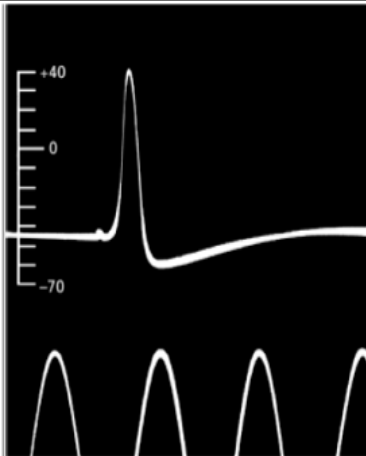
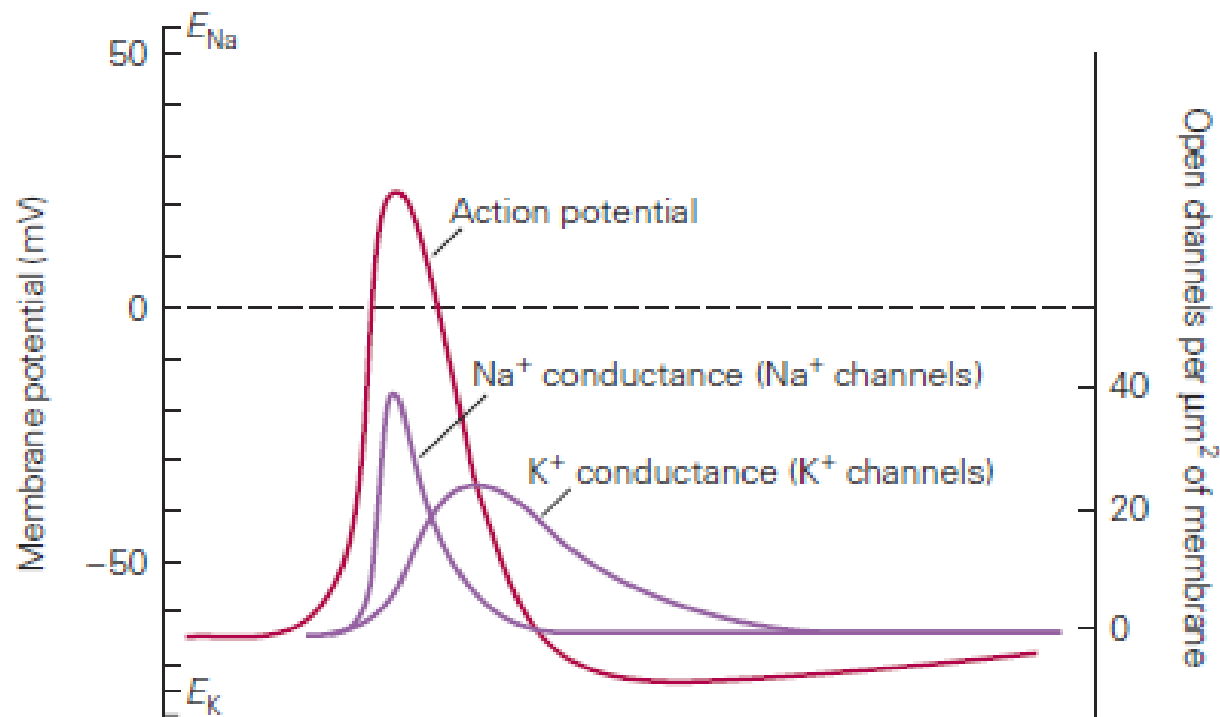
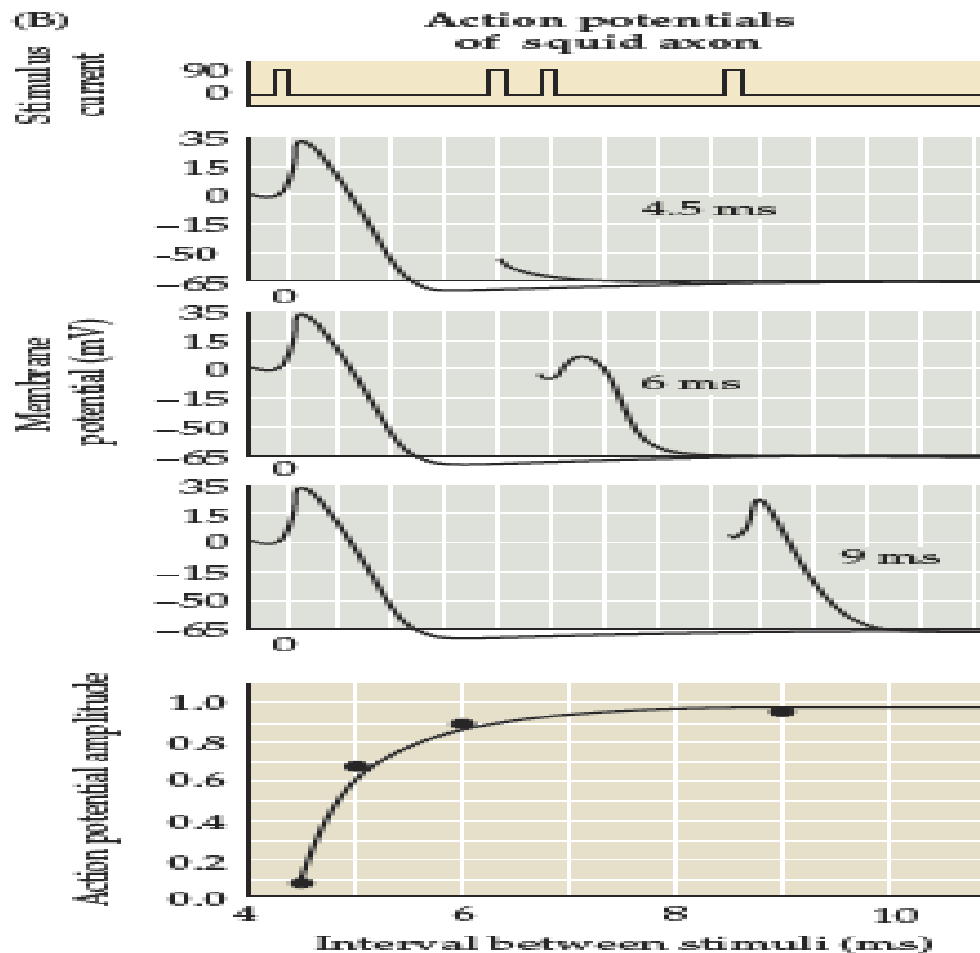


Figure 2-3 This historic tracing is the first published intracellular recording of an action potential. It was obtained in 1939 by Hodgkin and Huxley from the squid giant axon, using glass capillary electrodes filled with sea water. Time marker is 500 Hz. The vertical scale indicates the potential of the internal electrode in millivolts, the sea water outside being taken as zero potential. (From [Hodgkin and Huxley 1939](#).)





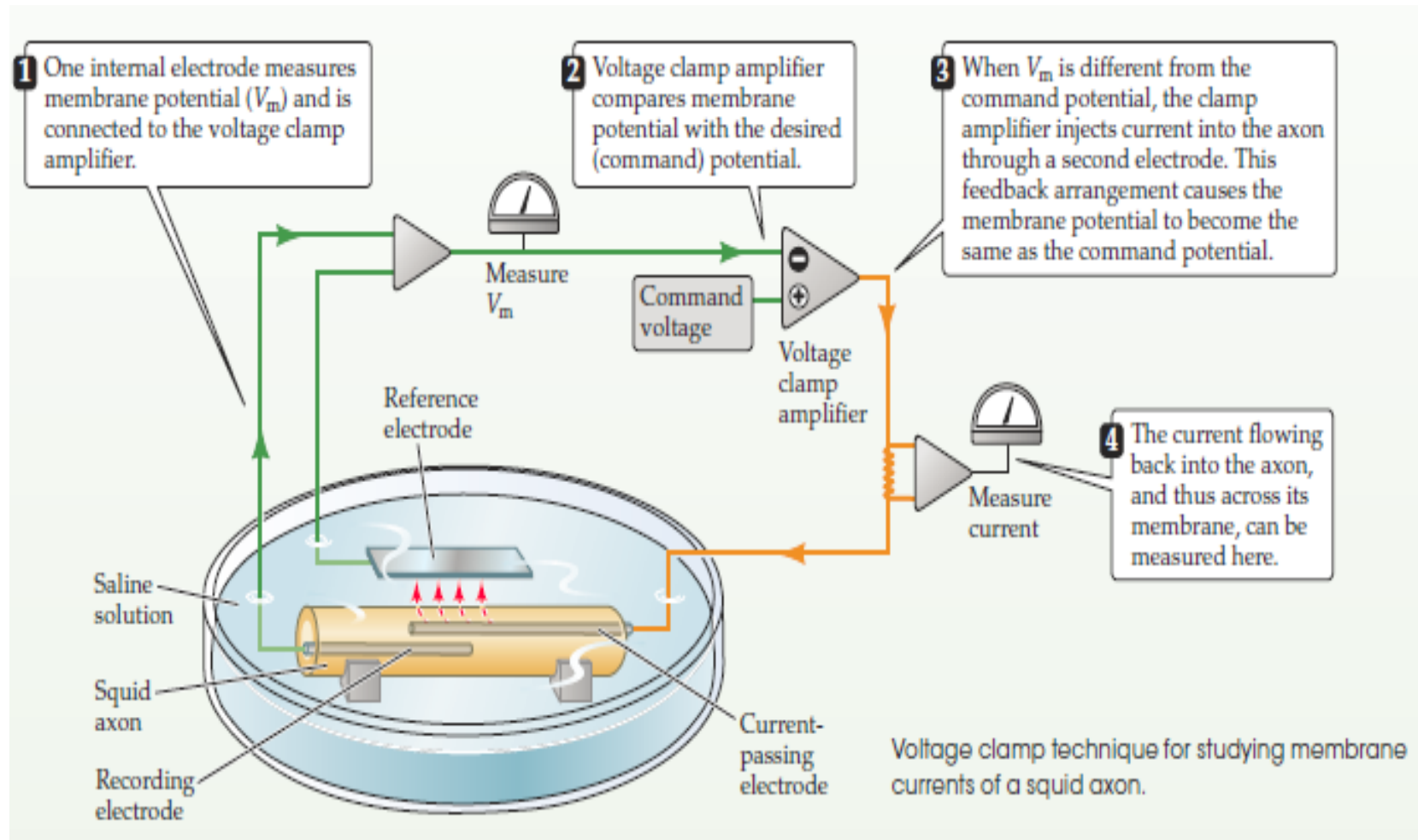
The sequential opening of voltage-gated Na^+ and K^+ channels generates the action potential

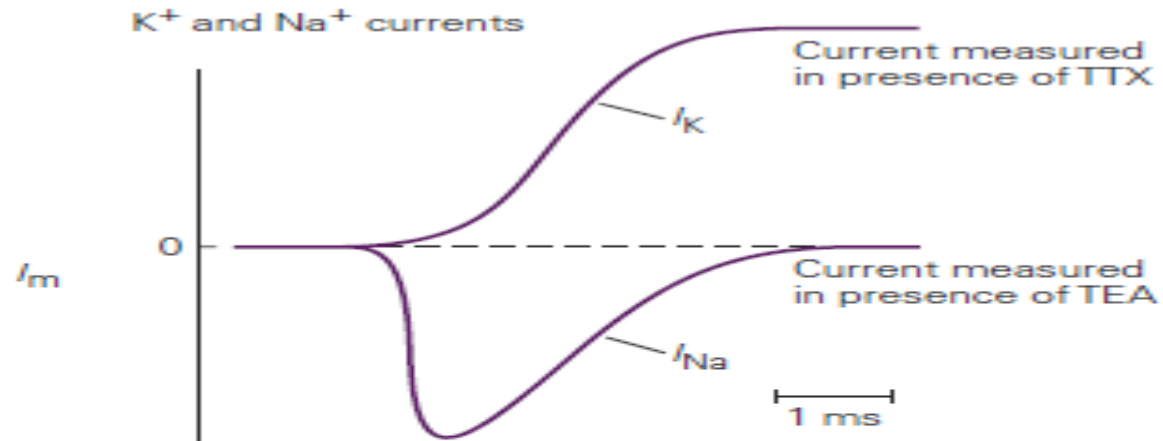
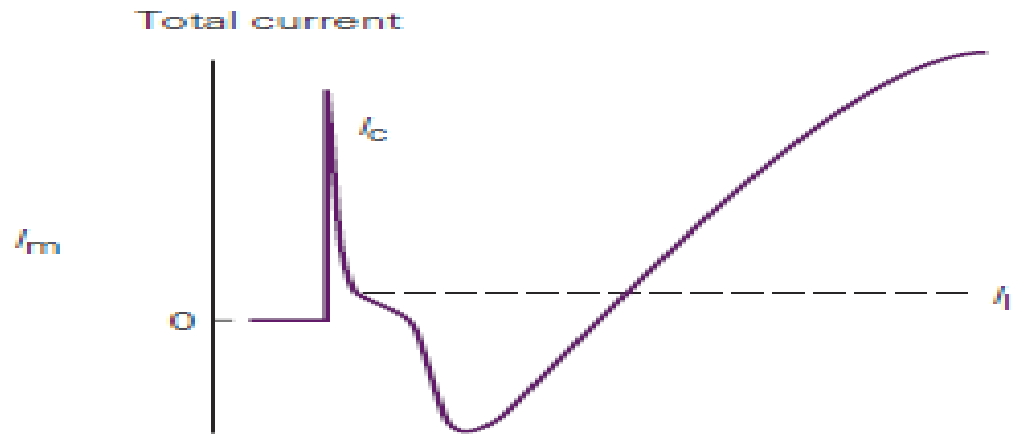


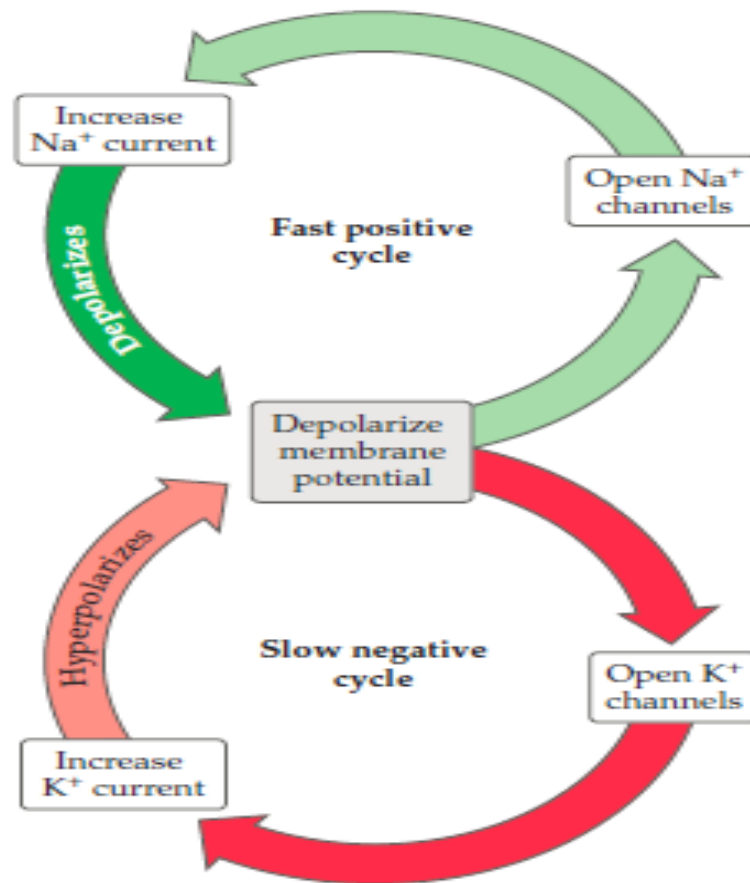
The refractory period can be observed by stimulating an axon with two current pulses that are separated by variable intervals. Whereas the first stimulus reliably evokes an action potential, during the refractory period the second stimulus will generate only a small action potential or no response at all.

IONIC MECHANISMS OF THE ACTION POTENTIAL

THE VOLTAGE CLAMP TECHNIQUE ALLOWS THE MEASUREMENT OF TRANSMEMBRANE CURRENTS

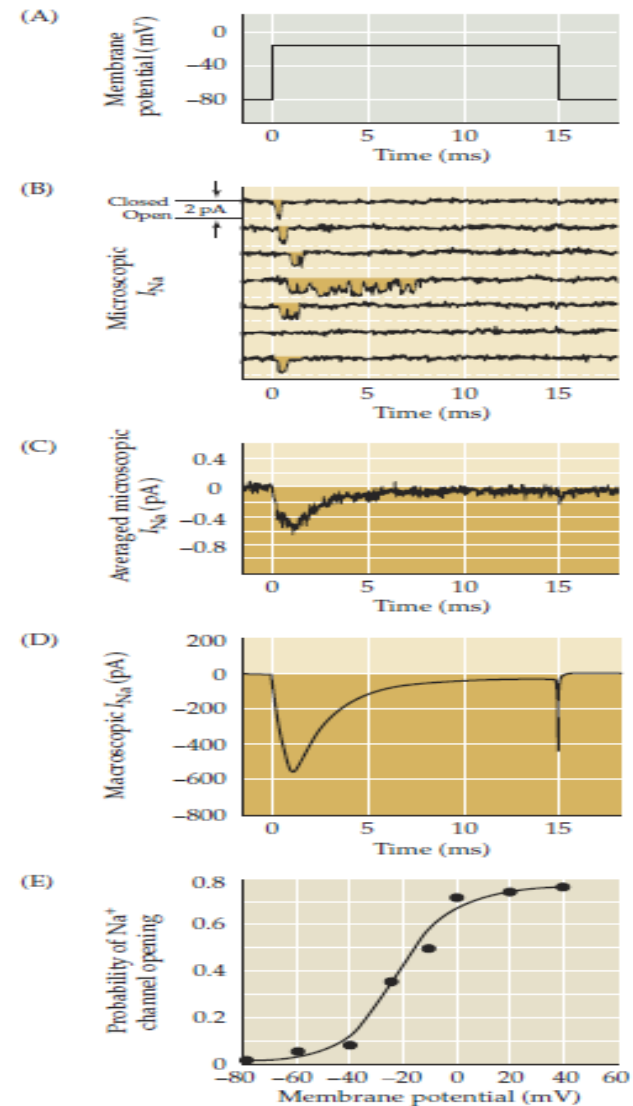
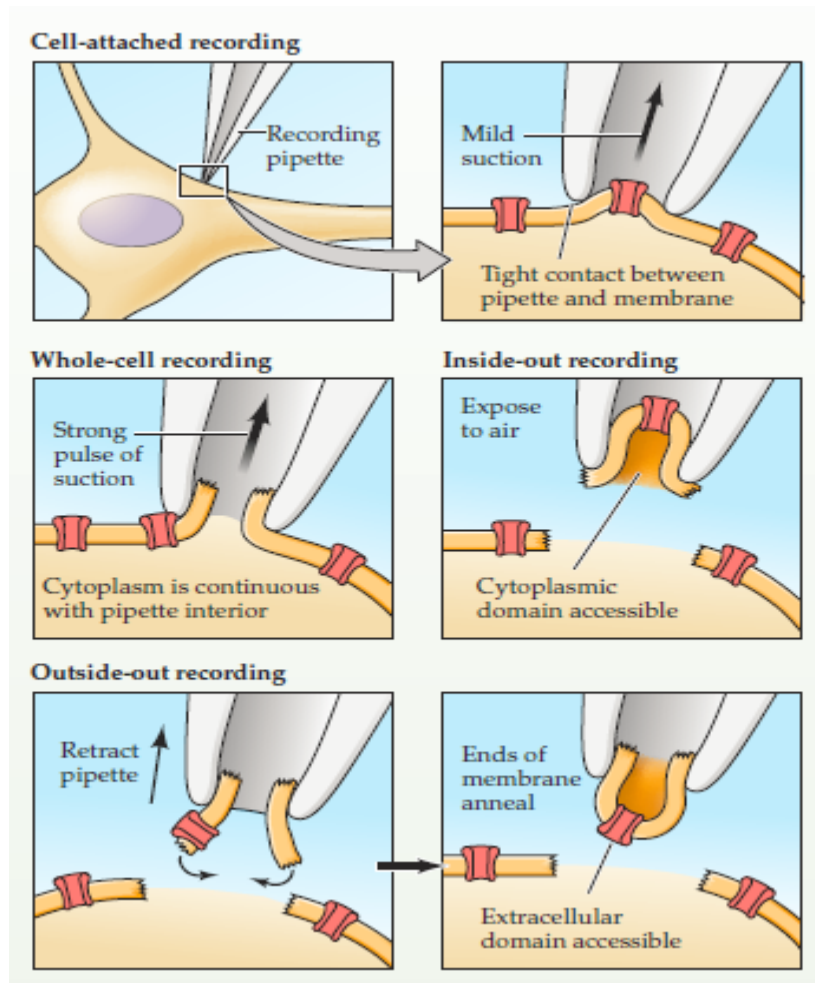




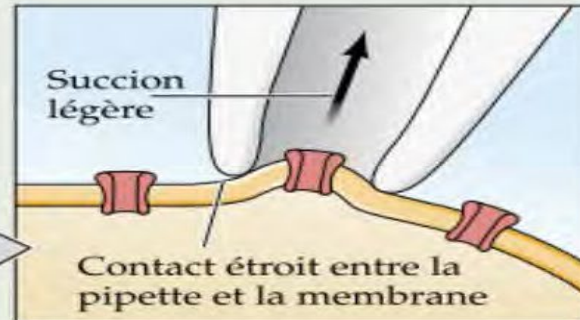
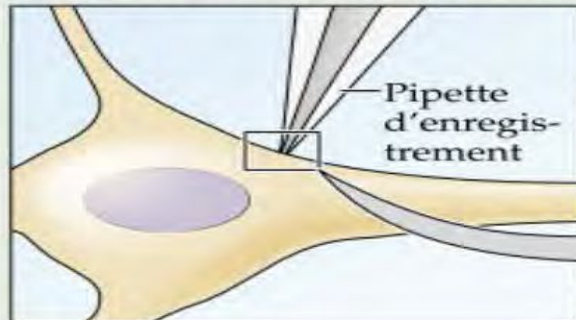


Feedback cycles are responsible for membrane potential changes during an action potential

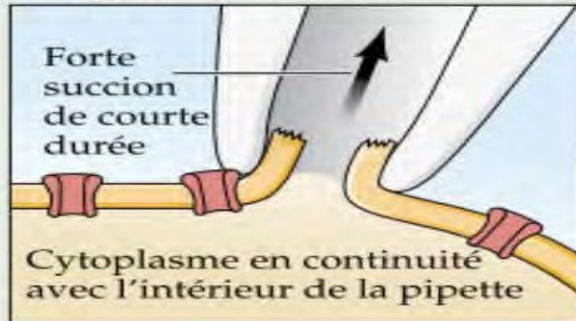
STUDY OF ION CHANNELS AT THE UNIT LEVEL: THE PATCH CLAMP TECHNIQUE



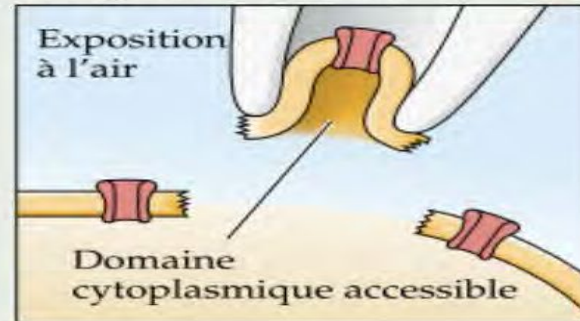
Enregistrement en cellule attachée



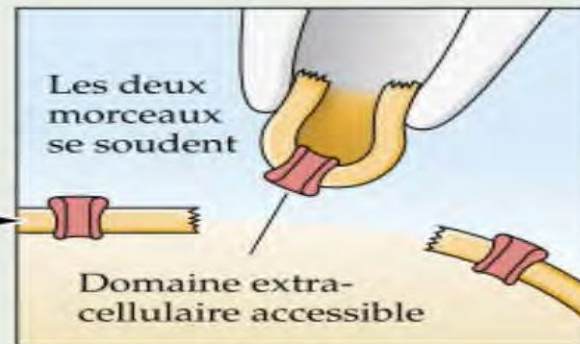
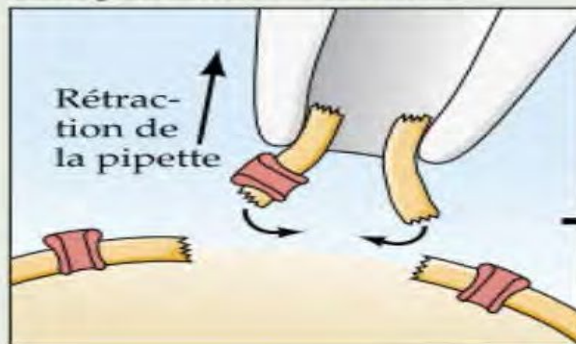
Enregistrement cellule entière



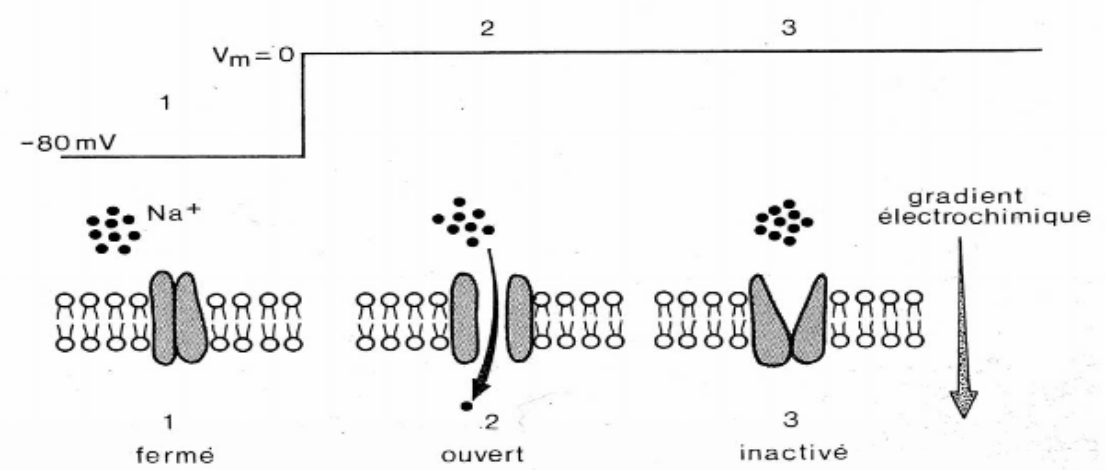
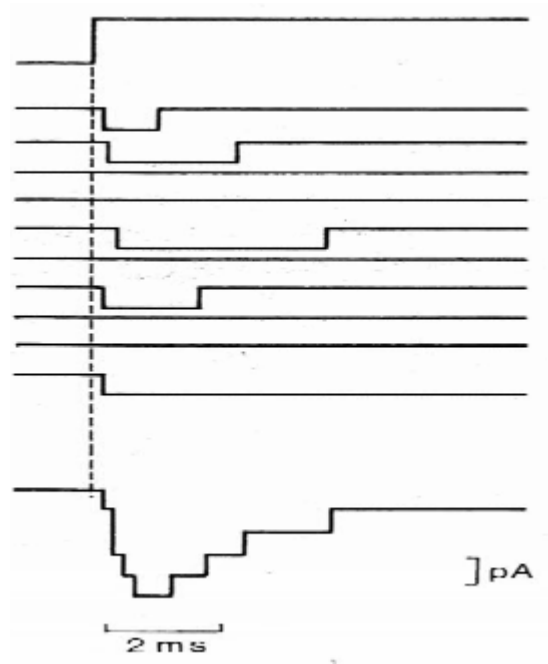
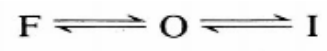
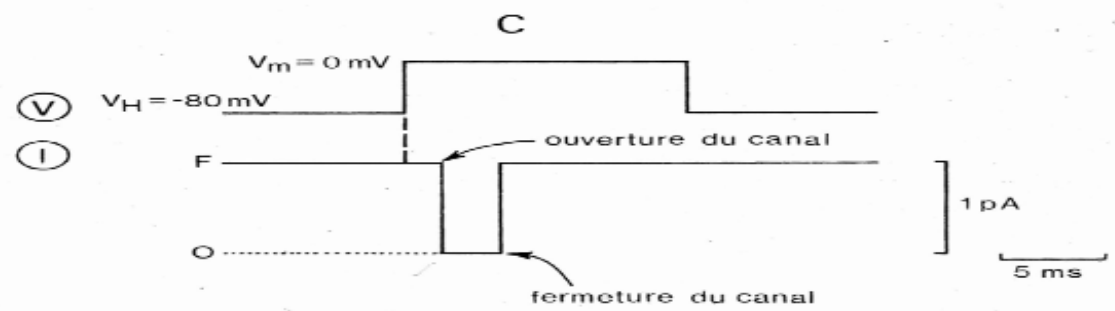
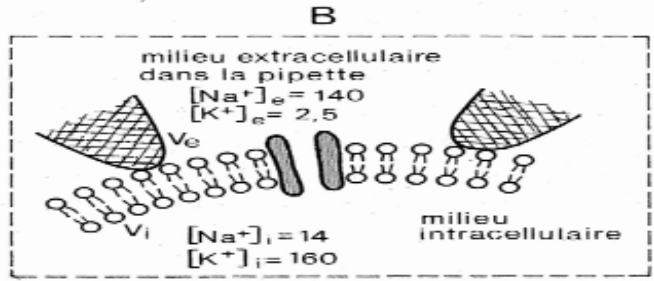
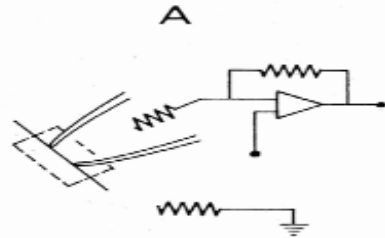
Enregistrement *inside out*

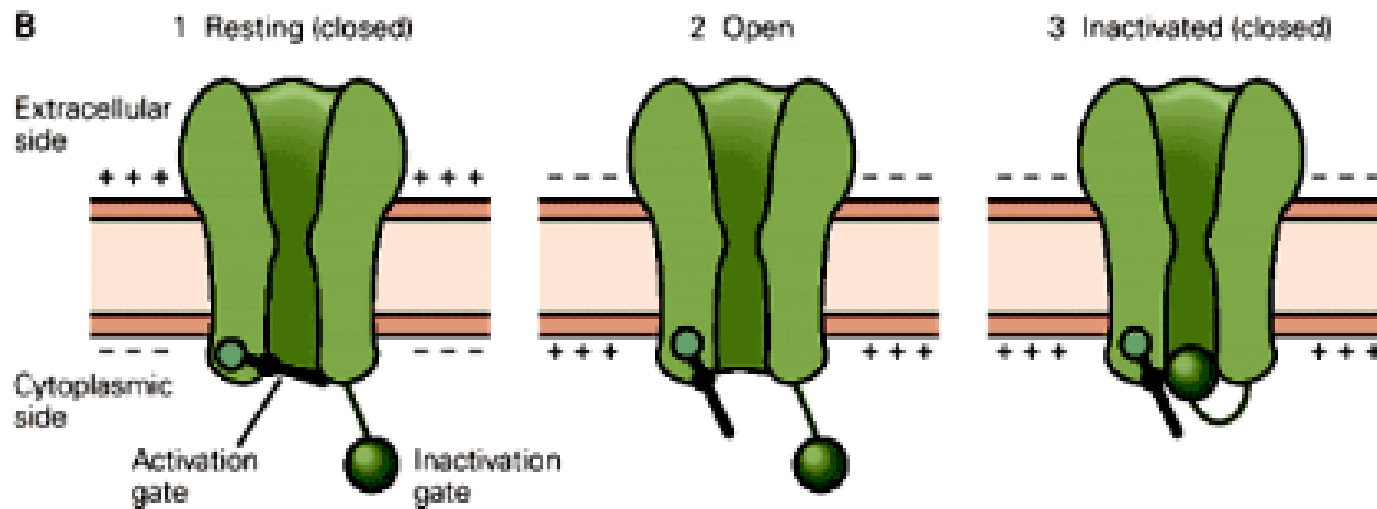
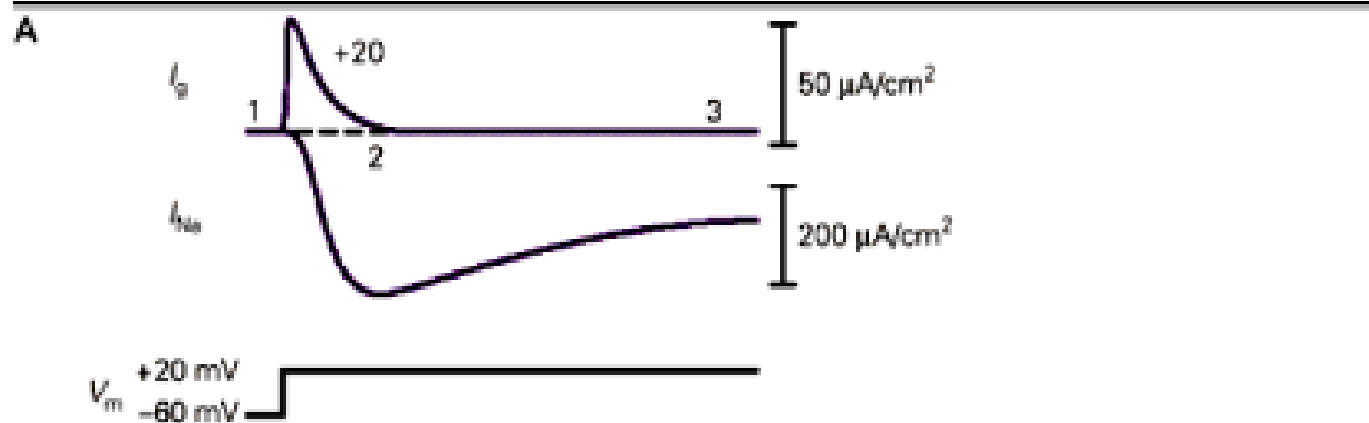


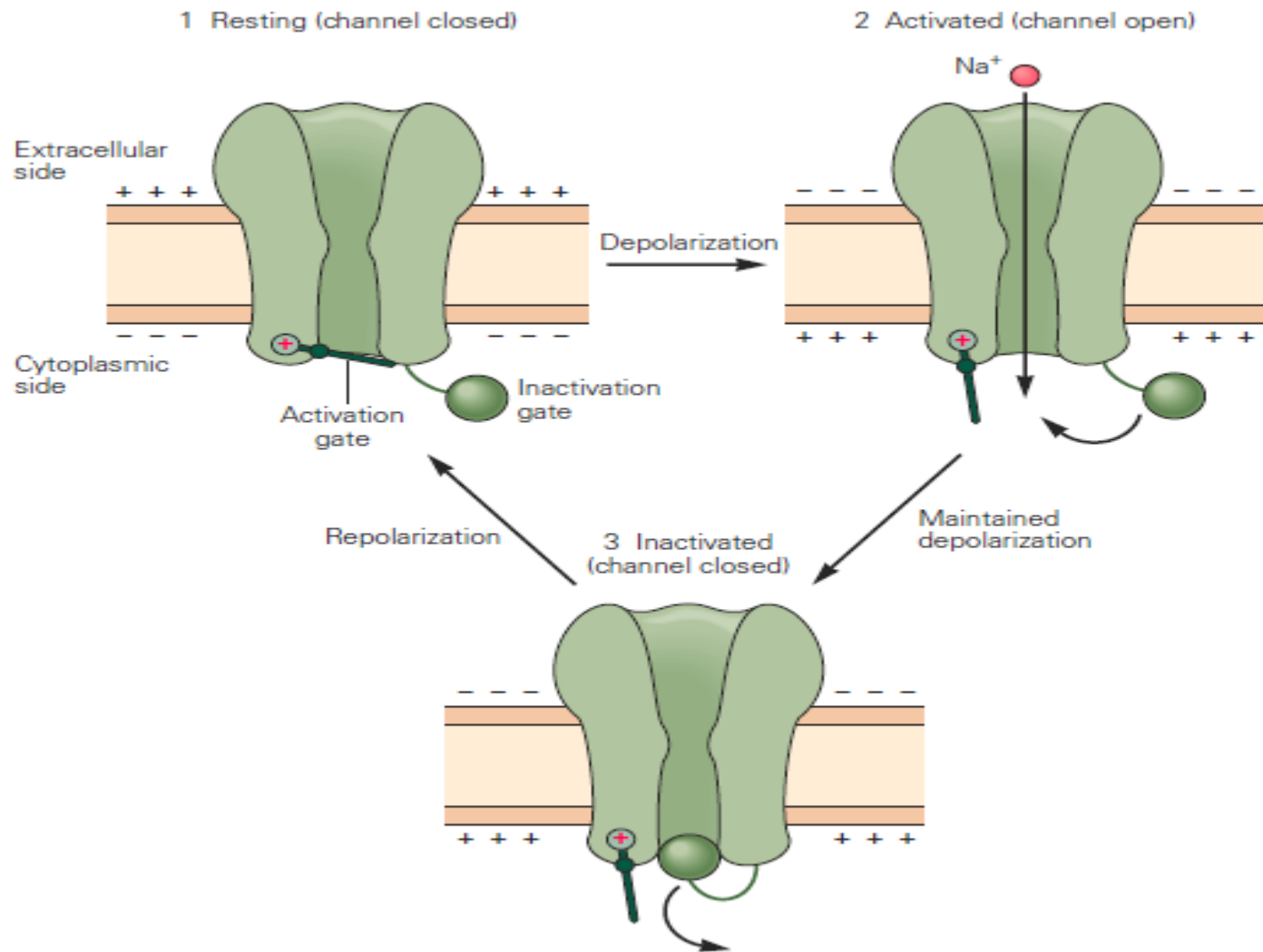
Enregistrement *outside out*

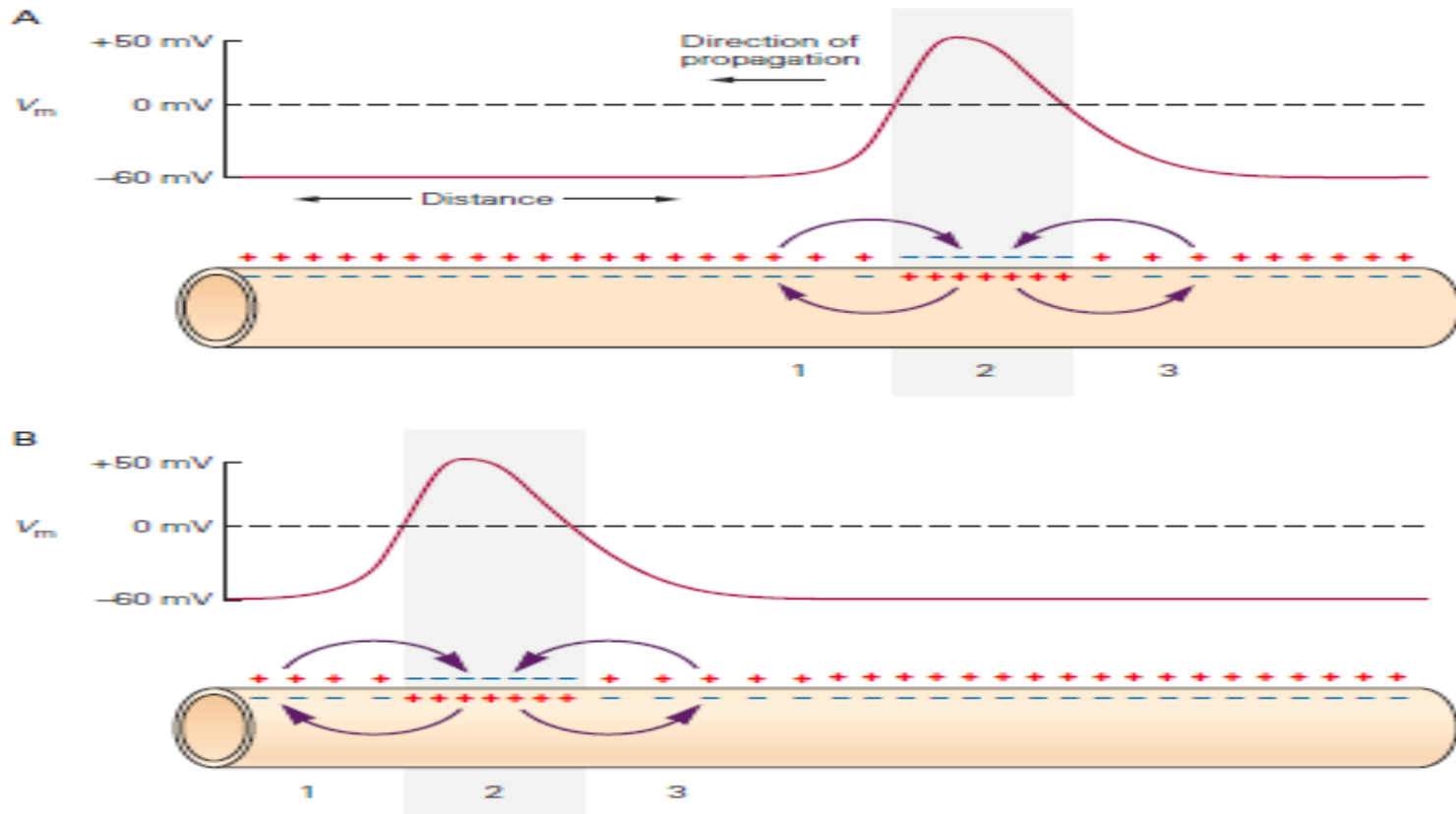


Quatre configurations pour la mesure en patch-clamp des courants ioniques



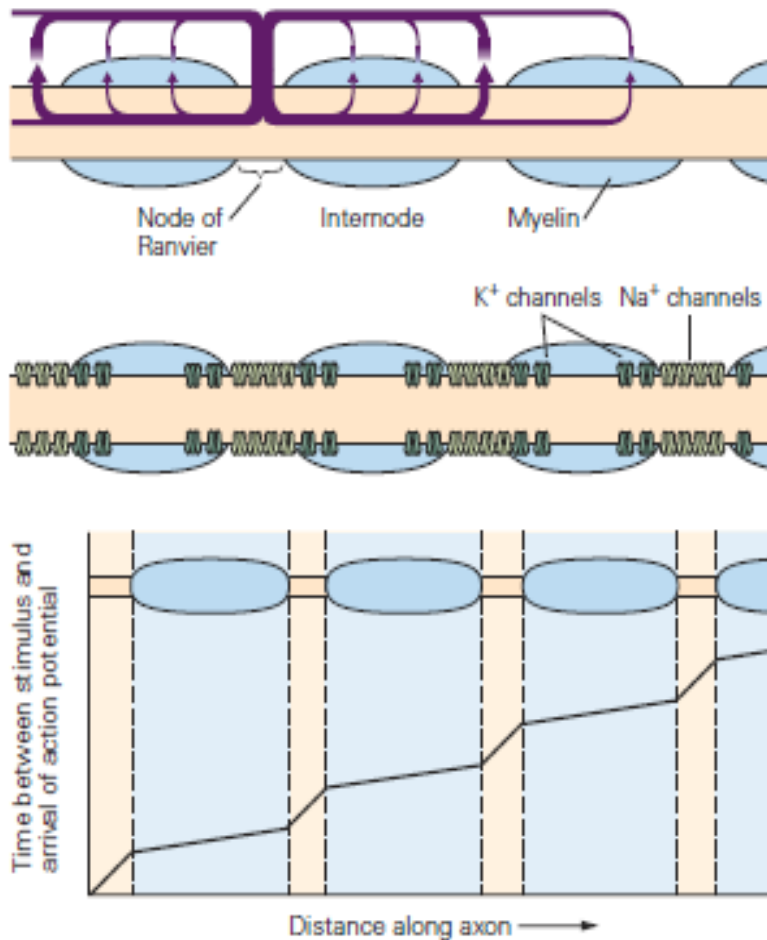




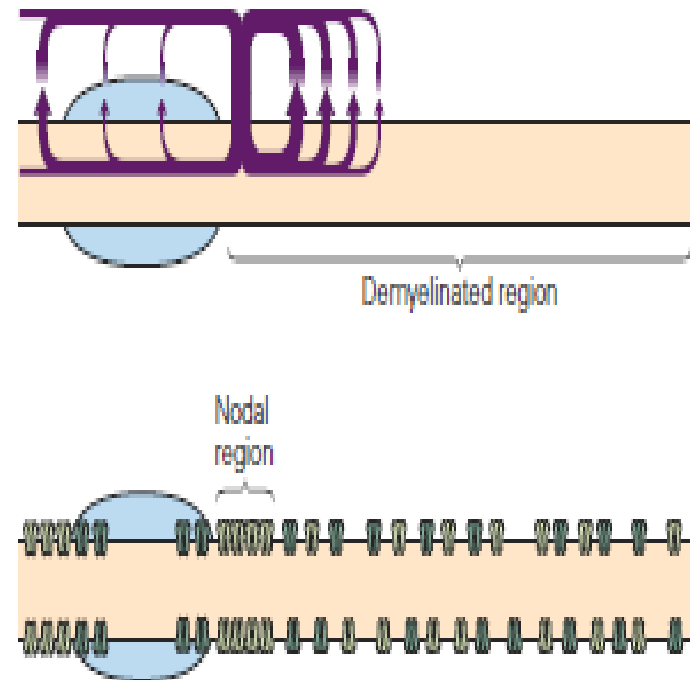


An action potential propagating from right to left causes a difference in membrane potential in two adjacent regions of the axon. The difference creates a **local-circuit current** that causes the depolarization to spread passively. Current spreads from the more positive active region (2) to the less positive resting region ahead of the action potential (1), as well as to the less positive area behind the action potential (3)

A Normal axon



B Demyelinated axon



Action potentials in myelinated nerves are regenerated at the nodes of Ranvier.